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M.Sc. Thesis

Multi-Task Symptom Learning and Explainability in Language Models

Classification of Depression

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Abstract

Computational natural language processing has shown great potential in automated depression classification. However, traditional binary classification models often act as uninterpretable black boxes. They frequently suffer from feature collapse, learning simple linguistic heuristics rather than gaining a clinically grounded understanding of depression. To address this, we introduce Symptom-based Multi-task Attention Representation Transformer (SMART), a BERT-based architecture that replaces standard binary classification practices with symptom-specific attention. Instead of directly optimising for a binary diagnosis, the model is trained to predict the eight symptom severity scores of the Patient Health Questionnaire-8 framework.

Empirical evaluation on the Distress Analysis Interview Corpus - Wizard of Oz demonstrates that this symptom-based approach successfully prevents feature collapse and significantly outperforms single-task baselines. The model achieves an accuracy of 0.80, a macro F1-score of 0.78, and a recall of 0.81 for the depressed minority class. Ablation experiments confirm that this predictive performance is driven by the symptom regression task, acting as a structural regulariser during fine-tuning.

To validate the clinical groundedness of the SMART model, we apply an explainability framework. Uniform Manifold Approximation and Projection reveals that the architecture structures patient embeddings along a continuous severity gradient rather than discrete binary clusters. Furthermore, layer-wise probing for $\mathcal{V}$-usable information and Shapley Additive Explanations for feature attribution demonstrate that multi-task learning forces the network to abstract structural syntax and self-reference in its deepest hidden layers, actively preventing shortcut learning. Finally, local token analyses highlight the limitations of unimodal text-only models, such as the modality gap for somatic symptoms and the semantic over-generalisation of physical vocabulary. Ultimately, this thesis demonstrates that symptom-level learning objectives provide a highly effective, interpretable alternative to binary black-box classification in computational psychology.

Contents

Chapter 1

Introduction

Mental Health disorders represent one of the most significant challenges to public health in recent years [Collaborators et al., 2022]. Unlike many physical illnesses that can be identified through clear biological markers such as blood tests or imaging, psychiatric diagnosis still largely relies on clinical observation [Smith et al., 2013] and subjective self-reporting [Kroenke et al., 2001]. However, language provides unique insights into the human mind. The way we communicate, choose words, structure our sentences, and organise the logical flow of our thoughts serves as a central marker of underlying cognitive and emotional states [Pennebaker et al., 2003].

Motivation

In mental disorders like depression, patients often show characteristic linguistic patterns. These patterns include increased self-focus, less complex sentence structures, or unexpected semantic associations [Rude et al., 2004, Al-Mosaiwi and Johnstone, 2018]. While clinicians can often detect these nuances, the process remains time-consuming and is prone to human bias [Blease et al., 2020].

Large Language Models (LLMs) such as Bidirectional Encoder Representations from Transformers (BERT) and RoBERTa have demonstrated an impressive ability to automatically classify mental disorders with high accuracy by interpreting large amounts of text [Ji et al., 2022]. However, the majority of the current text-based approaches suffer from a structural problem: They usually treat depression diagnosis as a simple binary task (depressed or healthy). Optimising only for a binary target can lead to feature collapse. The models act as uninterpretable black boxes that can make accurate predictions but find linguistic proxies and heuristics rather than building a clinically grounded understanding of the actual depression markers [Geirhos et al., 2020, McCoy et al., 2019, Ernala et al., 2019]. In a clinical setting, this lack of transparency and groundedness is essentially a barrier to trust. For AI to be integrated into psychiatric practice, we must move beyond pure binary prediction. We need to understand whether a model's internal logic aligns with established psycholinguistic theories or simply relies on low-quality correlations in the data.

Clinical Phenotypes and their Linguistic Manifestations

To analyse mental health using Natural Language Processing (NLP), we must first define the clinical characteristics of the target condition. This thesis focuses exclusively on Major Depressive Disorder (MDD). MDD is not a monolithic diagnosis. It is characterised by a diverse set of symptoms that manifest differently across patients. According to clinical frameworks such as the Patient Health Questionnaire-8 (PHQ-8) [Kroenke et al., 2009], these symptoms include:

- **Affective Symptoms:** Persistent low mood and loss of interest.
- **Somatic Symptoms:** Significant changes in sleep behaviour, appetite, and energy levels.
- **Cognitive Symptoms:** Feelings of worthlessness and low self-esteem, reduced ability to concentrate, and psychomotor agitation or retardation.

Psycholinguistically, these symptoms are often reflected in a linguistic shift towards higher self-reference, absolutist thinking, and a decrease in outgoing social engagement.

Problem Statement and Objectives

This thesis aims to address the limitations of standard binary classifiers by proposing a multi-task learning architecture called Symptom-based Multi-task Attention Representation Transformer (SMART). Instead of directly predicting a global diagnosis, we investigate whether the distinct symptoms, shown in the PHQ-8 questionnaire, can be detected individually from clinical interview transcripts.

The primary goal is to explore how forcing a language model to predict symptom severity scores changes its internal representation of clinical language. To achieve this, we evaluate the classification and regression performance of our proposed architecture on the Distress Analysis Interview Corpus - Wizard of Oz (DAIC-WOZ) dataset and perform explainability analyses using three methods:

- **Latent Space Analysis:** We visualise the geometric relationship of patient embeddings using Uniform Manifold Approximation and Projection (UMAP) to evaluate whether the model builds discrete binary clusters or a continuous severity gradient.
- **Feature Attribution:** We use Shapley Additive Explanations (SHAP) on a global and local level to highlight which specific tokens drive the predictions of the individual symptom heads and to identify potential contextual errors.
- **Layer-wise Probing:** We investigate across which layers the model encodes disorder-relevant features by predicting cognitive proxies such as surprisal, self-reference, and absolutist words.

Expected Contributions

We aim for this thesis to make both methodological and theoretical contributions to computational psychiatry. First, we demonstrate the performance of the SMART architecture and how multi-task symptom learning acts as a structural regulariser that successfully prevents feature collapse in clinical domains.

Secondly, we offer insights into model explainability by exploring which psycholinguistic features drive the diagnosis of depression. By investigating the model's layers and attention heads, we show how the network transitions from encoding linguistic heuristics to abstracting them into high-level clinical representations, while also addressing challenges like semantic overgeneralisation. This is particularly relevant for the development of Explainable AI (XAI) models that prioritise clinical accountability over opaque prediction.

Finally, this thesis explores the intersection of AI and psychology by proving that a symptom-based architecture structures the language internally in a more clinically grounded way. It shows that multi-task symptom learning can create internal representations that directly reflect the distinct cognitive and somatic markers of depression.

Thesis Outline

We first establish the required concepts and theoretical background used in this thesis in Chapter 2. We follow with an overview of the related work in Chapter 3. Chapter 4 details the methodology, including the dataset, the SMART architecture, and the explainability setup. The experimental results are presented in Chapter 5 and later interpreted in the Discussion in Chapter 6. Finally, Chapter 7 outlines the limitations of our approach and suggests directions for future work, before concluding the thesis in Chapter 8.

Chapter 2

Fundamentals

This chapter introduces the theoretical and technical foundation required for the methodology and experiments of this thesis. First, Section 2 introduces the field of computational psychiatry and the concept of language as a digital biomarker. Section 2 then details the evolution of neural language representations, focusing on the Transformer architecture and BERT. Section 2 explains the principles of multi-task learning and how it can be useful as a structural regulariser. Next, Section 2 introduces latent space analysis to understand how the model structures learned knowledge. We address the black-box problem in machine learning models and look at SHAP feature attribution and layer-wise probing in Section 2. Finally, Section 2 describes the psycholinguistic cognitive proxies used to evaluate the model's clinical groundedness.

Computational Psychiatry

The psychiatric diagnosis of mental disorders has historically been a challenge because it lacks clear physical markers like blood tests or X-rays [Insel, 2010]. Instead, professionals must rely on observing behaviour and, most importantly, investigating the patient's speech. Such a manual process remains subjective. It depends on the clinician's experience and the patient's ability to recall and articulate their symptoms during an interview.

Computational Psychiatry tries to move from these qualitative descriptions to more objective, mathematical models of mental disorders [Montague et al., 2012]. By treating language not just as a medium of communication, but as a measurable output of brain functions, we can use NLP to detect linguistic markers that are too subtle for a human clinician to catch.

Language as a Biomarker

Language is one of the most relevant outputs of the human brain [Bedi et al., 2015]. In clinical settings, speech and text are increasingly treated as digital biomarkers, i.e., objective, quantifiable indicators of internal cognitive and emotional states. We can use the way someone structures sentences, chooses vocabulary, or repeats certain pronouns to gain insights into their mental health [Spruit et al., 2022].

NLP focuses on developing algorithms that can automatically detect these markers. The transition from manual linguistic analysis to automated systems enables the processing of large data corpora, identifying subtle statistical patterns that may be invisible during standard human investigation [Low et al., 2020].

The Black-Box Problem

While AI models often achieve high performance, their decision-making process is completely non-transparent. In a medical setting, this black-box property is a major problem for real-world applications. A clinician can not simply trust a model's output stating that a patient is 85% likely to be depressed without understanding the reasoning behind it.

In this thesis, we therefore move beyond building a pure classifier. By investigating feature attribution and internal representations, we aim to learn whether the model's internal logic aligns with clinical understanding or relies on simple textual heuristics.

Neural Language Representations

From Static to Contextual Embeddings

The evolution of NLP is characterised by a transition from discrete to continuous and context-aware representations of linguistic units [Bengio et al., 2003]. Early approaches relied on one-hot encoding, where each word in a vocabulary V was represented as a sparse vector of size $|V|$ [Schütze et al., 2008]. This approach, however, falls short in capturing semantic relationships. The reason is that the dot product of any two distinct one-hot vectors is zero. This implies that semantically related words like *car* and *truck* are as distant from each other as unrelated ones like *car* and *flower*. Additionally, these sparse representations suffer from the curse of dimensionality [Bengio et al., 2003], making them computationally inefficient for large-scale applications.

The introduction of static word embeddings marked a shift towards distributed representations based on the distributional hypothesis, which proposes that words occurring in similar contexts share similar meanings [Harris, 1954]. Models such as Word2Vec [Mikolov et al., 2013] and GloVe

[Pennington et al., 2014] map tokens into a continuous, low-dimensional vector space, denoted by $\mathbb{R}^d$. In this latent space, semantic similarity is reflected by geometric properties, typically calculated using cosine similarity:

$$similarity(u, v) = \frac{u \cdot v}{\|u\|\|v\|}$$

While static embeddings capture lexical relationships through vector arithmetic [Mikolov et al., 2013], they are fundamentally limited by the polysemy problem [Camacho-Collados and Pilehvar, 2018]. Since each word is assigned a single, fixed vector, these models cannot distinguish between different senses of the same token [Camacho-Collados and Pilehvar, 2018]. For instance, the word *depressed* would have an identical representation in both a clinical and an economic context, despite the significant differences in their semantic and diagnostic implications. To overcome these limitations, contextualised word representations were developed to produce embeddings that adapt to the surrounding text. Early architectures such as ELMo utilised bidirectional Long Short-Term Memory networks to incorporate context [Peters et al., 2018], but were significantly outperformed by the Transformer-based approach of BERT [Devlin et al., 2019]. Unlike static models, contextualised encoders generate a sequence of vectors $(h_1, h_2, \ldots, h_n)$ for a given input $(t_1, t_2, \ldots, t_n$, where each vector is a function of the entire sequence:

$$h_i = f_\theta(t_i | t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_n)$$

In this framework, the representation of a token is computed by attending to the entire sequence simultaneously [Vaswani et al., 2017]. In the context of clinical applications, this allows the model to differentiate meaning based on specific linguistic markers, such as vocabulary or affect, which is essential for accurate classification within the domain of mental health [Spruit et al., 2022].

The Transformer Architecture and Self-Attention

The Transformer architecture is the core innovation that enables contextualised representations by replacing recurrence with an attention mechanism [Vaswani et al., 2017]. Recurrent Neural Networks process tokens sequentially, which limits their ability to capture long-range dependencies due to the vanishing gradient problem [Hochreiter, 1998]. In contrast, the Transformer processes the entire input sequence in parallel.

The Self-Attention Mechanism

At the heart of the Transformers encoder is the attention mechanism. For each token, the model computes three functional vectors: a Query (Q), a key (K), and a Value (V). These vectors are derived by multiplying the input embedding x by the learned weight matrices $W^Q, W^K, W^V \in \mathbb{R}^{d \times d_k}$. To understand the intuition behind this mechanism, the interaction between Query, Key, and Value can be compared to a retrieval system. The Query represents the current token looking for relevant information, the Key acts as a descriptor of what other tokens in the sequence can offer, and the Value contains the information that is passed forward once a match is found. The attention mechanism determines the importance of other tokens in the sequence relative to a target token by computing a compatibility function between the query and each key. The output is a weighted sum of the values, where the weights are assigned by a softmax function:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

The scaling factor $\sqrt{d_k}$, where d_k is the dimension of the keys, is applied to prevent the magnitude of the dot product from growing too large, which would result in the softmax function being pushed into regions with extremely small gradients [Vaswani et al., 2017].

Multi-Head Attention

While a single attention function can capture relationships between tokens, it is often limited by its ability to focus on multiple types of linguistic information simultaneously. As a solution, the Transformer uses Multi-head attention, which allows it to attend to information from different representation subspaces at different positions [Vaswani et al., 2017]. Rather than performing a single attention operation, the model computes a linear projection of the queries, key, and values h times, where h is the number of attention heads, utilising distinct learned weight matrices for each head. Each head then performs the attention operation in parallel:

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$$

The independent outputs of these heads are concatenated and projected back to the final model dimension d_model using a weight matrix W^O:

$$MultiHead(Q, K, V) = Concat(head_1, \ldots, head_h)W^O$$

In the context of mental health classification, this architecture is especially beneficial. It allows the model to resolve complex linguistic dependencies by dedicating different heads to distinct features. Empirical findings indicate that some attention heads address syntactic structures while others focus on semantic affect or emotional valence [Clark et al., 2019]. Viewing a text from these multiple angles is essential for identifying subtle markers of mental disorders, which often occur as a combination of syntactic rigidity and shifted semantic focus.

Bidirectional Encoder Representations from Transformers

In this thesis, we use BERT, a multi-layer bidirectional transformer encoder. Its architecture is based on the original transformer implementation [Vaswani et al., 2017], but excludes the decoder stack, as the primary goal is to generate embeddings rather than text [Devlin et al., 2019]. Depending on the model size, the architecture consists of a certain number of encoder layers, hidden size units, and attention heads.

Input Representation

To process text, BERT uses a stepwise embedding mechanism. Before an input sequence gets processed by the first transformer layer, it is transformed into a sum of three distinct embeddings:

- **Token Embeddings:** The raw input text is tokenised using the WordPiece algorithm [Wu et al., 2016], which breaks words into word pieces. This effectively minimises the risk of encountering out-of-vocabulary words, which is important when dealing with the atypical language found in clinical data.
- **Segment Embeddings:** These embeddings distinguish between sentences within a single input by indicating which sentence each token belongs to.
- **Position Embeddings:** Since the model has no inherent sense of word order, these embeddings provide the model with the relative or absolute position of each token in the sequence.

The final input representation for a token is the element-wise sum of these three embeddings (figure).

Pre-Training Objectives

BERT is pre-trained using large corpora of unlabeled data. It is optimised using two primary objectives:

- **Masked Language Modeling:** Unlike the conventional approach of predicting the next word in a sequence, BERT simply masks a percentage of words from the input tokens at random and then learns to predict those marked tokens. This forces the model to develop a symmetrical understanding of language where the representation of a token is influenced by the entire surrounding sequence. Mathematically, BERT minimises the cross-entropy loss:

$$L_{MLM} = -\sum_{i \in M} logP(x_i|\tilde{x}),$$

where M are the masked positions of the input sequence, x_i is the token at position i that we want to recover, and $\tilde{x}$ is the input sequence where tokens at positions in M have been replaced. In this thesis, this allows us to extract the likelihood of specific patient utterances. A word that is highly surprising to the model often indicates unnatural cognitive processing or absolutist thinking, which are key markers of depression.

- **Next Sentence Prediction:** The model is trained to predict whether a given sentence B logically follows sentence A. This objective helps BERT capture sentence relationships, which is a crucial feature for understanding the overall conversational flow and contextual coherence in clinical interviews.

Bidirectionality

Unlike previous bidirectional models like ELMo [Peters et al., 2018], which simply concatenate two separate unidirectional layers, one reading from left to right and the other reading from right to left, BERT uses a single transformer stack that allows every token to attend to its entire surrounding context simultaneously across all layers. This becomes especially important in the context of this master's thesis, as in the psychiatric context, the meaning of a word is often ambiguous until the entire sentence is processed.

The [CLS] Token

In order to allow tasks like classification, BERT introduces a dedicated architectural element called the [CLS] token. The primary purpose of this token is to serve as a fixed-position representation that captures the aggregated semantic information of the entire input sequence. The model is designed to pool information from all subsequent tokens into a single vector through the self-attention mechanism by prepending [CLS] to every text sample.

During the encoding process, as the text passes through the transformer layers, the [CLS] token interacts with every other word in the sequence. Consequently, the final hidden state of the network, known as the [CLS] embedding, serves as a summary of the input.

In the context of a classification task, a classification head is directly attached to this embedding. This head typically consists of a simple linear layer that maps the [CLS] embedding to the desired output, using a function such as softmax or sigmoid to obtain class probabilities:

$$P = softmax(W \cdot [CLS] + b).$$

While standard classifiers rely heavily on this [CLS] embedding to make a final prediction, this thesis explores an alternative multi-task attention mechanism to extract symptom-specific representations. However, the [CLS] token remains highly valuable for analysis. Because it attends to every word during the encoding phase, it passively collects the overall clinical context of the input. In this thesis, we extract these [CLS] embeddings to project and visualise the latent space, allowing us to analyse how the model internally structures depression severity.

Multi-Task Learning

Traditionally, machine learning models are designed to solve a single task by optimising a single loss function. Multi-task learning, however, is a practice where a model is trained to solve multiple related tasks simultaneously. This approach improves generalisation performance by forcing the model to find a shared representation that satisfies all objectives at once [Caruana, 1997]. Multi-task learning is commonly implemented through hard parameter sharing. In this architecture, the network shares its foundational layers, such as a common BERT encoder, while keeping task-specific output layers. This thesis utilises hard parameter sharing by using a single transformer backbone paired with multiple regression heads and a classification head.

Structural Regulariser

Another advantage of multi-task learning is the introduction of an inductive bias, acting as a structural regulariser. In single-task learning, a model trained on a small or highly specific dataset risks overfitting to random statistical noise that happens to correlate with the single target variable.

By contrast, a multi-task learning architecture must learn features that are universally useful across all task objectives. Features that only benefit a single task are naturally suppressed during training because they would increase the loss of the other tasks. Consequently, the network is forced to learn a more general representation of the input text.

Mitigating Feature Collapse

The regularisation effect in multi-task learning is particularly valuable for clinical NLP, which often deals with imbalanced datasets. When standard single-task models are trained on imbalanced data to predict a binary diagnosis, they can suffer from feature collapse [Papyan et al., 2020, Benton et al., 2017]. Instead of learning clinically grounded markers, the model finds mathematical exploits, such as always predicting the majority class to minimise the loss.

Multi-task learning can prevent this limitation because a model can not rely on a single binary heuristic. To minimise the combined loss function, the network has to explicitly learn the distinct dimensions of the targeted condition, resulting in a more robust and clinically grounded reasoning.

Latent Space Analysis

In deep learning and NLP, models map input data into high-dimensional vector representations. Within this latent space, semantically similar concepts are positioned closer together, while unrelated concepts are pushed apart (cite). Understanding the structure of this space is crucial for interpreting how a neural network conceptually encodes information.

Since we cannot visualise these high-dimensional embeddings directly, we use a manifold learning technique to project the embeddings into a two- or three-dimensional space to make them interpretable.

A popular dimensionality reduction method is UMAP [McInnes et al., 2018]. UMAP is particularly good at preserving both the local and global structure of the data. This means that the relative distances and topological distributions in the 2D plot are an accurate reflection of the actual geometry in the high-dimensional latent space. This allows for visually investigating whether a model has successfully learned meaningful clusters or continuous gradients from the underlying data.

Explainability and Interpretability in NLP

LLMs like BERT are increasingly deployed in sensitive domains [Harrer, 2023] such as clinical psychology. One of the biggest challenges is the black-box nature of these architectures [Rudin, 2019]. While high predictive accuracy is desirable, it does not ensure that the model's decisions are based on clinically relevant features but merely on dataset statistics or simple heuristics. Interpretability of these models refers to the degree to which a human can understand the reasoning of a decision, whereas XAI involves the technical methods used to make the internal mechanisms of a model transparent and understandable to humans [Adadi and Berrada, 2018].

Shapley Additive Explanations

SHAP is an attribution method based on cooperative game theory, namely on the concept of Shapley values [Lundberg and Lee, 2017]. In NLP, each token in an input sequence is treated as a player in a game, and the model's prediction is the outcome. The goal is to distribute the outcome fairly among the tokens according to their individual contribution.

Formally, we consider a set of input features $F = \{1, 2, \ldots, p\}$, where any given subset $S \subseteq F$ forms a coalition of features that jointly contribute to a specific model outcome $val(S)$. To ensure a fair predictive baseline, we assume $val(\emptyset) = 0$. The core logic of Shapley values lies in redistributing the total predictive outcome, $val(F)$, by determining the average marginal contribution of each feature across every possible coalition. The marginal contribution of a single feature i relative to a coalition S is defined as:

$$\Delta val(i, S) = val(S \cup \{i\}) - val(S)$$

This formula allows us to quantify the specific value added by introducing feature i to an existing set of features. This contribution is averaged over all possible subsets $S \subseteq F \setminus \{i\}$ that do not contain the feature. The final corresponding Shapley value $\phi_{val}(i)$ is calculated with:

$$\phi_{val}(i) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(p - |S| - 1)!}{p!} \Delta val(i, S).$$

The coefficient $\frac{|S|!(p-|S|-1)!}{p!}$ acts as a normalisation term that accounts for the different permutations of the subset S.

Using SHAP in our experiments allows us to identify relevant linguistic markers by considering all possible feature interactions. This is crucial in the context of multi-task and symptom-based depression classification, as the significance of a word might only emerge through its relationship with other terms in the patient's speech, allowing us to evaluate whether the individual symptom heads operate based on clinical logic.

Layer-wise Probing

While attribution methods like SHAP highlight which tokens are responsible for a prediction, they still treat the model's internal mechanisms as a black box. To investigate how linguistic and cognitive information is structured and passed across the different layers of the network, we use layer-wise probing [Belinkov, 2022].

In this framework, the internal hidden state of a particular layer l in the neural network is treated as a representation $R^{(l)}$ that potentially encodes various properties. To test if a specific target property Y, such as sentiment or cognitive surprisal, is present, a probing classifier, also called a probe, is introduced. This probe is a small supervised model, usually consisting of a single linear layer, that attempts to perform a probing task $R^{(l)} \to Y$, mapping from the frozen representations to the target property.

If this classifier can successfully extract the information Y from the representation $R^{(l)}$ with a high accuracy, this serves as evidence that the model has internally encoded that specific feature Y at that layer l [Alain and Bengio, 2016]. One might wonder why the use of such a simple linear probe is logical, however, this decision is crucial in ensuring that the information is already present and easily accessible within the representation rather than being computed by the probing classifier itself in a complicated and non-linear way.

$\mathcal{V}$-usable Information

Some studies suggest that simple test set accuracy may be an insufficient metric for evaluating how well features are encoded in different layers. Accuracy can be heavily influenced by the fitting power of the probe itself, or fail to distinguish between layers once performance approaches 100%. To provide a more faithful measurement, we want to adopt the concept of $\mathcal{V}$-usable information or $\mathcal{V}$-information in short [Xu et al., 2020].

$\mathcal{V}$-usable information measures how easy it is for a specific model family $\mathcal{V}$ to extract information. In our case, $\mathcal{V}$ is the family of linear classifiers. The $\mathcal{V}$ information is mathematically defined as the difference between the predictive $\mathcal{V}$-Entropy $H_{\mathcal{V}}(Y)$ of the labels and the conditional $\mathcal{V}$-Entropy $H_{\mathcal{V}}(Y|R^{(l)})$ given the hidden representations. The final formula is given by:

$$I_{\mathcal{V}}(R^{(l)} \to Y) = H_{\mathcal{V}}(Y) - H_{\mathcal{V}}(Y|R^{(l)}),$$

where $H_{\mathcal{V}}(Y)$ and $H_{\mathcal{V}}(Y|R^{(l)})$ can be estimated using:

$$\begin{aligned} H_{\mathcal{V}}(Y) &= \inf_{f \in \mathcal{V}} \mathbb{E}[-\log_2 f[\emptyset](Y)], \\ H_{\mathcal{V}}(Y|R^{(l)}) &= \inf_{f \in \mathcal{V}} \mathbb{E}[-\log_2 f[R^{(l)}](Y)], \end{aligned}$$

with $\emptyset$ denoting that in the case of no information, we use zero-input vectors as an alternative representation.

This metric allows us to better differentiate how knowledge is passed through the model's layers. In this thesis, computing the $\mathcal{V}$-usable information reveals if and at which exact architectural depth the network encodes complex psycholinguistic markers, such as self-reference and absolutist words.

Cognitive Proxies in NLP

As described in the previous chapter, layer-wise probing enables us to gain a better understanding of the information stored within the different layers of our BERT architecture, however, the choice of meaningful target properties is crucial. In this thesis, we explore cognitive proxies that serve as established indicators of underlying psychological states or cognitive impairments in MDD [Chancellor and De Choudhury, 2020].

By investigating these proxies, we can evaluate whether the model's internal representations capture the same signals that are utilised for clinical diagnosis. This allows us not only to perform a technical evaluation of the interpretability analysis but also to conduct a theory-driven validation of the model's clinical groundedness.

Specifically, we examine how SMART encodes information related to cognitive surprisal, self-reference, and absolutist thinking, as these are known to be linguistic markers for depression [Al-Mosaiwi and Johnstone, 2018, Rude et al., 2004].

Surprisal

One of the central proxies we will explore is surprisal, which is derived from Information Theory [Shannon, 1948, Levy, 2008]. In a linguistic context, surprisal quantifies how unexpected a word is given its preceding context. Mathematically, for a token w_i, the surprisal $I(w_i)$ is defined as the negative log-probability of the token given its prior sequence:

$$I(w_i) = -\log P(w_i|w_1, \ldots, w_{i-1}).$$

In current computational linguistics, the surprisal can be estimated using pre-trained autoregressive language models like GPT-2. Because the models are structurally forced to predict the next word sequentially, their cross-entropy loss serves as a direct measurement of word surprisal.

This metric correlates strongly with human cognitive processing effort. Words that are highly unpredictable in a given context require greater cognitive resources to integrate [Levy, 2008]. In the context of depression, surprisal is a relevant proxy for cognitive depletion. Depressed patients often show a reduced diversity in vocabulary and repetitive thought loops, resulting in a more predictable linguistic structure [Smirnova[1] et al., 2013]. By probing our model layers for surprisal, we can evaluate whether the network learns and encodes the structural predictability of a patient's speech as a digital biomarker for symptoms of depression.

Self-Reference

Another cognitive proxy we investigate is the self-reference bias. This linguistic marker denotes an increased focus on the self, which manifests as a significantly higher frequency of first-person singular pronouns such as *I*, *me*, and *mine* [Zimmermann et al., 2017]. [Rude et al., 2004] suggests that pronoun density is often a more reliable predictor of depressive symptoms in mental disorders than the use of negative emotion words.

Absolutist Words

The final cognitive proxy we examine is absolutist thinking. Depressed patients often demonstrate cognitive distortions such as black-and-white thinking. Linguistically, this is shown through a higher frequency of absolutist words. These are terms that indicate absolute magnitudes, such as *always*, *never*, or *completely*.

Studies have shown that a high frequency of absolutist words in language is a highly distinct marker for anxiety and depression

[Al-Mosaiwi and Johnstone, 2018]. In this thesis, investigating how the model encodes absolutist words is especially important. Because the multi-task architecture is designed to predict symptom severity scores, the network must learn to evaluate the intensity of a patient's statements. Probing for absolutist words allows us to verify whether the model correctly identifies this linguistic intensity rather than just classifying based on general context.

Chapter 3

Related Work

Automated depression detection using clinical language has been an active area of research in recent years. Most studies attempting to predict MDD from patient speech are based on the DAIC-WOZ dataset [Gratch et al., 2014]. Because this dataset provides audio, video, and text transcriptions, a large portion of previous work approached depression classification as a multimodal problem. These models typically combine acoustic features like voice signals with visual indicators [Williamson et al., 2016, Koops et al., 2023].

In this thesis, however, we focus on text as a single modality to isolate linguistic and cognitive markers. Textual data captures the semantics of a patient's speech without adding possibly noisy audio or video signals. Therefore, we mostly limit our review to studies that have also focused primarily on the text transcriptions of clinical interviews. To provide an overview, Section 3 reviews the architectural approaches used for this task. Next, Section 3 explores the advances and challenges in symptom-level classification, and finally, Section 3 highlights the critical need for explainability in clinical NLP.

Architectural Approaches

Initial research on modelling clinical interview transcripts focused on capturing the sequential properties of patient language. Early deep learning approaches applied to the DAIC-WOZ dataset mostly relied on Recurrent Neural Networks such as Long Short-Term Memory networks or Gated Recurrent Units to extract textual features over time [Al Hanai et al., 2018, Ringeval et al., 2019]. Recently, the focus has shifted to Transformer architectures. Researchers started using pre-trained models like BERT, RoBERTa, and domain-specific variants like Clinical BERT [Alsentzer et al., 2019] or MentalBERT [Ji et al., 2022] to get better contextualised embeddings for clinical language.

However, since long interviews produce transcripts that are above the allowed token limit in Transformers, researchers have to find effective ways to solve this limitation. For example, [Gavalan et al., 2024] proposed a text summarisation approach for the DAIC-WOZ dataset. They used KeyBERT [Grootendorst, 2020] to extract the most informative keywords, effectively compressing the interviews before passing them through a CNN together with a custom depression lexicon. While their summarisation successfully solves the token limit problem and reaches a macro F1-score of 0.67, filtering only for keywords is not ideal. It risks deleting the natural conversational flow and subtle linguistic signals, which are essential diagnostic markers for depression.

Beyond addressing technical constraints like sequence length, another research area focuses on feeding models with domain-specific knowledge. Some works explicitly modelled the affective features of words using psycholinguistic lexicons like LIWC to improve classification accuracy [Trotzek et al., 2018]. Other papers built multi-task architectures to predict the DAIC-WOZ binary diagnoses alongside emotion from external datasets [Li et al., 2022].

Symptom-level Classification

Almost all of these previous studies on clinical interview data developed models to predict a single binary diagnosis or a single severity score [Haque et al., 2018]. Symptom-level prediction on clinical datasets is still rarely explored. While a paper by [Delahunty et al., 2019] used the DAIC-WOZ dataset to model the comorbidity between depression and anxiety, their focus was limited to predicting only two main depression symptoms rather than the full profile. [Milintsevich et al., 2023] expanded on this approach by actively modelling the severity of all eight PHQ-8 symptoms. By treating the problem as a multi-target regression task, they showed that predicting individual symptom scores can result in a more interpretable evaluation than a single binary output.

Despite this progress, modelling multiple symptoms simultaneously introduces a new problem. When multi-task learning is applied to clinical data, models often suffer from feature collapse. Instead of learning specific linguistic markers for each symptom dimension, the model simply calculates a generalised prediction of negative affect to minimise its loss. The challenge of designing a multi-task architecture that successfully prevents this collapse is still not fully solved.

Explainable Model Architectures

While predictive accuracy on these models has improved, recent studies question the validity of models trained on the DAIC-WOZ dataset. [Burdisso et al., 2024, Patapati et al., 2025] demonstrated that many high-performance classifiers do not actually learn the linguistic markers of depression. Instead, they overfit to dataset-specific statistics like length and frequency of the virtual interviewer's questions.

This problem highlighted the need for XAI systems in clinical NLP. Most research uses attribution methods like SHAP or LIME [Lundberg and Lee, 2017, Ribeiro et al., 2016] to visualise which input tokens are relevant in model decisions. While these tools effectively show which words a model looks at, they do not explain what clinical concepts the network actually understands. Psycholinguistic research demonstrated that depression is often characterised by deeper structural patterns such as the frequency of first-person singular pronouns or shifts in linguistic predictability [Zimmermann et al., 2017, Smirnova[1] et al., 2013]. Simply highlighting individual tokens does not prove whether a model has truly learned these underlying cognitive markers. To verify this, the model's internal representations must be examined directly.

Our thesis aims to explore these identified gaps. By introducing a symptom-specific attention mechanism combined with a correlation penalty, we want to improve multi-task symptom regression while avoiding feature collapse. To handle the sequence length limit without losing conversational context, we use a chunking and patient-level aggregation approach rather than text summarisation. Furthermore, by employing layer-wise probing for cognitive markers like absolutist words and surprisal, we aim to verify whether the model genuinely learns clinical knowledge or purely relies on statistics in the dataset.

Chapter 4

Methodology

First, Section 4 introduces the DAIC-WOZ dataset, followed by the required data preparation steps in Section 4. Section 4 presents the core architecture of this thesis: a multi-task learning framework designed for symptom-score prediction. The corresponding training procedure and the custom loss function are described in Section 4. To properly evaluate our approach, Section 4 defines the baseline models and Section 4 lists the evaluation metrics. Finally, Section 4 outlines the experimental setup for the interpretability methods used to validate the models.

Distress Analysis Interview Corpus - Wizard of Oz

We trained and evaluated our models on the extended version of the DAIC-WOZ dataset. This dataset was initially introduced by [Gratch et al., 2014] and later expanded as part of the AVEC 2019 workshop and challenge [Ringeval et al., 2019]. The extended dataset contains audio and video files for a total of 275 clinical interviews together with their transcripts, designed to capture psychological distress. The interviews were either conducted by a virtual interviewer called "Ellie", who was human-controlled or by a fully autonomous AI. All participants were adult Americans, primarily from the West Coast (Los Angeles, California), with 168 patients being male, 105 being female, and 2 not having a specified gender. The duration of the recorded interviews varies based on the participant's responsiveness, ranging from around 8 minutes to 31 minutes.

The interviews are accompanied by the PHQ-8 [Kroenke et al., 2009], which is used for assessing the depression severity of the patients. The questionnaire consists of eight distinct questions, each corresponding to a specific diagnostic symptom of depression. In this thesis, we call the symptoms referred to by these questions Interest, Mood, Sleep, Energy, Appetite, Self-esteem, Concentration, and Psychomotor. Participants self-report the frequency of these symptoms using a 4-point scale (0 = not at all, 1 = several days, 2 = more than half the days, 3 = nearly every day). The sum of these eight questions produces a total severity score from 0 to 24. In the dataset, patients are defined as depressed if their total severity score is greater than or equal to 10.

Finally, the dataset has a predefined separation into train, test, and validation splits, and we applied some filtering to the dataset, removing the sessions that lack the official PHQ-8 scores or are unusable due to problems with the audio. The exact distributions of the dataset with their binary labels can be seen in Table 4.1, and the distribution of the PHQ-8 severity scores can be found in Table 4.2.

Table 4.1: Binary label distribution across the DAIC-WOZ dataset splits.

Dataset Split	Total Patients	Healthy	Depressed
Train	154	118 (76.6%)	36 (23.4%)
Validation	53	41 (77.4%)	12 (22.6%)
Test	50	34 (68.0%)	16 (32.0%)
Total	**257**	**193 (75.1%)**	**64 (24.9%)**

Preprocessing

To train our model effectively, the raw transcripts from the DAIC-WOZ dataset require a preprocessing pipeline. We want to extract meaningful language while discarding uninformative noise.

Each patient interview was segmented into multiple discrete .txt files representing individual conversational turns or segmented timeframes of the patient's speech, removing the interviewer's parts. Instead of concatenating these files into a single transcript, which could exceed BERT's token input limit, each text file was registered as an independent data sample. This segmentation approach naturally expands the dataset to sufficient training instances, providing the model with a diverse set of short and focused clinical statements from which to learn specific symptom markers.

In clinical interviews, a large portion of the dialogue consists of filler words or brief acknowledgements (e.g., *yes*, *no*, *Mhm*) [Spruit et al., 2022]. Even though these natural parts of human language could be used as linguistic patterns, such as speech hesitation or markers of cognitive load, they lack the semantic density required for our neural network to detect depressive symptoms. Therefore, we used a filter that discards any text segments containing fewer than five words. This step is crucial and ensures that our model is trained on deeper psycholinguistic patterns rather than shallow conversational artifacts.

Table 4.2: Distribution of the PHQ-8 symptom scores for each datasplit (in %).

Symptom	Score	Train	Dev	Test	Total
Interest	0	50.0%	41.5%	44.0%	47.1%
	1	34.4%	43.4%	32.0%	35.8%
	2	12.3%	7.5%	14.0%	11.7%
	3	3.2%	7.5%	10.0%	5.4%
Mood	0	42.9%	43.4%	34.0%	41.2%
	1	37.0%	35.8%	38.0%	37.0%
	2	11.7%	13.2%	20.0%	13.6%
	3	8.4%	7.5%	8.0%	8.2%
Sleep	0	41.6%	41.5%	32.0%	39.7%
	1	24.0%	28.3%	24.0%	24.9%
	2	17.5%	15.1%	20.0%	17.5%
	3	16.9%	15.1%	24.0%	17.9%
Energy	0	31.8%	26.4%	30.0%	30.4%
	1	39.0%	43.4%	36.0%	39.3%
	2	16.2%	17.0%	14.0%	16.0%
	3	13.0%	13.2%	20.0%	14.4%
Appetite	0	47.4%	35.8%	32.0%	42.0%
	1	26.0%	37.7%	34.0%	30.0%
	2	14.3%	18.9%	18.0%	16.0%
	3	12.3%	7.5%	16.0%	12.1%
Self-esteem	0	48.1%	41.5%	42.0%	45.5%
	1	25.3%	34.0%	28.0%	27.6%
	2	14.9%	15.1%	20.0%	16.0%
	3	11.7%	9.4%	10.0%	10.9%
Concentration	0	56.5%	54.7%	40.0%	52.9%
	1	22.7%	24.5%	26.0%	23.7%
	2	9.1%	15.1%	22.0%	12.8%
	3	11.7%	5.7%	12.0%	10.5%
Psychomotor	0	77.9%	71.7%	64.0%	73.9%
	1	13.0%	20.8%	18.0%	15.6%
	2	5.2%	5.7%	14.0%	7.0%
	3	3.9%	1.9%	4.0%	3.5%

Because the PHQ-8 scores are assessed at the patient level and not the segment level, for each participant, we applied the eight individual symptom scores, the binary diagnosis label, and the total summed PHQ-8 score to all valid text segments belonging to that participant. While this creates some degree of noise (a patient with disturbed sleep patterns might not discuss this topic in every single segment), the amount of data allows our model to isolate the true linguistic correlations. The model learns to predict the continuous symptom vector for each text chunk, establishing the foundation for the subsequent patient-level aggregation phase during inference.

The Symptom-based Multi-task Attention Representation Transformer Architecture

To effectively classify depressive disorders from text, we propose a new model architecture and pipeline based on the pretrained MentalBERT model from HuggingFace [Ji et al., 2022]. While traditional BERT-based classifiers utilise a single linear classification layer attached to the final hidden state of the [CLS] token to provide an aggregate diagnostic prediction, our approach tries to preserve the clinical grounding of depressive disorders.

An overall visualisation of the proposed SMART architecture is shown in Figure 4.1

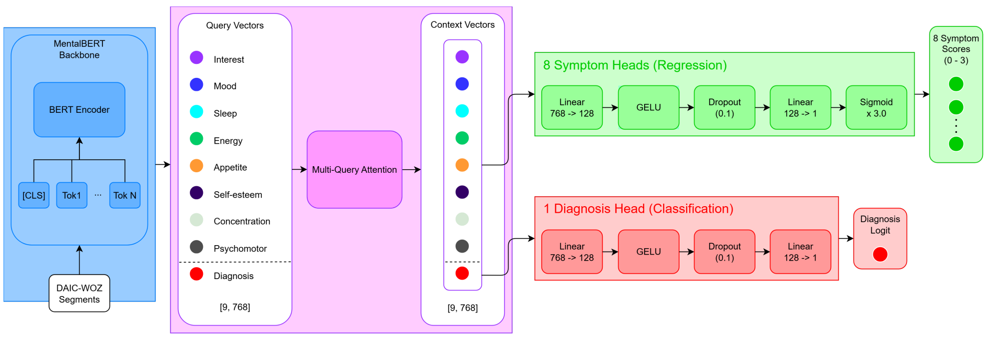

Figure 4.1: Architecture of the Symptom-based Multi-task Attention Representation Transformer (SMART). The input sequences from the DAIC-WOZ dataset are fed into the MentalBERT backbone to extract full sequence representations. A following multi-query attention mechanism calculates 9 context vectors, eight dedicated to the PHQ-8 symptoms and one for the binary diagnosis. Finally, these vectors are processed by independent multi-task production heads where the symptom heads output symptom severity scores and the diagnosis head outputs a diagnosis logit.

Input Encoding and MentalBERT Backbone

We fed our model with the preprocessed transcriptions of the patient's responses from the DAIC-WOZ clinical interviews. This input was processed by a pretrained MentalBERT backbone, a multi-layer bidirectional encoder loaded from HuggingFace. Unlike standard BERT applications that utilise only the pooled [CLS] output tokens for classification, our architecture extracts the full sequence hidden states $H \in \mathbb{R}^{T \times 768}$ from the final encoder layer, where T represents the number of tokens. These states provide a high-dimensional and context-rich representation of the patient's speech, serving as the semantic foundation for the symptom-specific analysis.

Symptom-Specific Multi-Query Attention

At the core of our architecture lies a specialised attention mechanism that replaces the standard global pooling layer for BERT classification models. We introduce nine learnable query vectors $Q = \{q_1, q_2, \cdots, q_9\}$, each with a dimension of 768:

- **Symptom Queries ($q_1, q_2, \cdots, q_8$):** Each vector is dedicated to identifying linguistic markers associated with one of the eight symptoms defined by the PHQ-8 framework, such as Sleep, Mood, or Psychomotor changes.
- **Diagnosis Query (q_9):** This vector is specialised in capturing the aggregate clinical features required for a global diagnostic assessment.

Each query q_i computes attention weights over the hidden states H, identifying the most relevant tokens for its specific clinical symptom in the PHQ-8. The result is a set of nine context vectors (1 x 768), each representing a targeted summary of the input segment tailored to a specific symptom or the overall diagnosis.

Multi-Task Prediction Heads

The generated context vectors are subsequently passed into independent multi-task heads. To prevent overfitting and capture non-linear relationships, each head is structured as a multi-layer perceptron. The 768-dimensional context vector is first projected into a 128-dimensional layer, followed by a GELU activation and dropout (p=0.1). A final layer maps this representation to a single numerical output. The architecture distinguishes between two types of predictive tasks:

- **Symptom Regression Heads:** The first eight context vectors are processed by individual linear layers mapping the 768-dimensional representation to a single scalar. These heads also apply a sigmoid activation multiplied by 3, leading to an output of a symptom score on a scale of 0 to 3. This represents the quantitative rating system used in the PHQ-8 framework to assess the frequency and severity of individual depressive symptoms and is supervised against the PHQ-8 scores given by the patient. During inference, these heads provide a symptom-level view for each text segment.

- **Diagnosis Classification Head:** The ninth context vector is processed by a dedicated classification head that outputs a diagnostic logit. This represents the model's global indicator for the likelihood of a depression diagnosis. However, this only serves as a multi-task learning tool for the model to better understand the symptom relationships with the overall diagnosis and is not used in the final model decision. The integration of this logit into the loss function is explained in Section 4.

From Segment-level to Session-level

A critical component of our pipeline is the transition from segment-level predictions to a session-level diagnosis. Since each clinical interview consists of multiple utterances, the model must aggregate information across the entire dialogue.

For a given interview session S consisting of n segments $\{s_1, s_2, \cdots, s_n\}$, the model first calculates the sum of the eight predicted symptom intensities for each segment. If $\hat{y}_{i,j}$ is the predicted regression score for symptom i in the segment j, the total score for segment j is defined as:

$$\hat{y}_j = \sum_{i=1}^{8} \hat{y}_{i,j}.$$

The final session-level PHQ-8 score is then calculated as the mean of all segment scores for that specific patient:

$$Score_{total} = \frac{1}{n} \sum_{j=1}^{n} \hat{y}_j.$$

Biased Threshold

To ensure that the final model decision is not just a black-box probability but allows for a high degree of interpretability, we used the sum of the predicted symptom scores $Score_{total}$ and compared it against a decision threshold θ. While it is suggested that a PHQ-8 score of 10 or more is likely to successfully classify a patient as depressed [Kroenke et al., 2009], our pipeline accounts for the specific distribution and inherent bias of the DAIC-WOZ dataset.

By analysing the distribution of predicted scores against the actual labels in the training set, we calculated an adjusted threshold that accounts for model-specific calibration and data prevalence. A patient is classified as depressed if:

$$Score_{total} + Bias \geq \theta.$$

This step is necessary for the clinical groundedness of our model, as it allows for the fine-tuning of sensitivity based on the actual diagnostic distribution of the dataset, ensuring that it aligns with the gold standard.

Training Procedure

A challenge we encountered during training is the simultaneous optimisation of distinct mathematical objectives while preventing feature collapse. To solve this problem, we implemented a custom loss function within a trainer class that integrates three specific penalty components:

- **Symptom Regression Penalty:** Our main objective is to accurately predict the continuous severity scores for the eight PHQ-8 symptoms. We achieved this by using the MSE between the predicted symptom scores and the clinical ground truth. For a batch size of N, the loss is defined as:

$$L_{MSE} = \frac{1}{N \cdot 8} \sum_{j=1}^{N} \sum_{i=1}^{8} (\hat{y}_{j,i} - y_{j,i})^2,$$

  where $\hat{y}_{j,i}$ is the predicted score for symptom i in batch j and $y_{j,i}$ is the clinical ground truth.

- **Diagnosis Classification Penalty:** We employed a binary cross-entropy with logits loss for the binary diagnosis prediction. To account for the bias in the number of healthy controls in the DAIC-WOZ dataset, we used a weight of $w_p = 3.0$, which is applied to the depressed class. This weight penalises the model strongly for predicting false negatives, prioritising clinical recall, which is most important in such clinical applications [Kuhn et al., 2013]. Finally, we integrated a sigmoid activation σ for numerical stability and calculated the loss for a batch size of N with:

$$L_{BCE} = -\frac{1}{N} \sum_{j=1}^{N} (w_p \cdot y_j \cdot \log(\sigma(\hat{x_j})) + (1 - y_j) \cdot \log(1 - \sigma(\hat{x_j}))),$$

  where $\hat{x_j}$ is the logit output for sample batch j and $y_j \in \{0, 1\}$ is the binary ground truth.

- **Correlation Penalty:** In initial experiments, we noticed that the eight symptom heads redundantly predicted the same global sentiment. Therefore, we introduced this correlation penalty. During the forward pass, the batch predictions for the symptoms are centred, normalised, and used to calculate a Pearson correlation matrix R. We ignored the self-correlations and only activated this penalty if the absolute correlation between any two distinct symptom predictions lay above a threshold of 0.7, which was determined through heuristic evaluation during experiments. Mathematically, the loss is defined as:

$$L_{correlation} = \frac{1}{56} \sum_{i \neq j} \max(0, |R_{i,j}| - 0.7).$$

This loss forces our model to learn independent representations for each symptom.

The final loss L_{total} during backpropagation is a weighted combination of these three components.

$$L_{total} = 0.7 \cdot L_{MSE} + 0.3 \cdot L_{BCE} + 0.3 \cdot L_{correlation}.$$

Since the primary goal of our model is to predict the eight symptom scores accurately, we assigned a higher coefficient to the MSE loss L_{MSE} than to the secondary objective loss L_{BCE}. All the coefficients have been determined through an empirical hyperparameter tuning based on the validation set performance.

The model was fine-tuned using the AdamW optimiser with a learning rate of $2 \cdot e^{-5}$ and a batch size of 16. The training was done for a maximum of 15 epochs using an early stopping mechanism with a patience of 5. We utilised the macro F1-score on the validation set for early stopping and automatically restored the best performing model weight at the end of the training process.

The pipeline was implemented using Python and the PyTorch framework. The full implementation code is available at GitHub [1]

Baseline Models

To thoroughly evaluate our proposed SMART architecture, its performance was compared against two baselines to isolate the benefits of task-specific fine-tuning and multi-task architectural complexity.

For a fair comparison, both baselines evaluated the texts from the dataset at a sentence level, aligned with our approach for the SMART model. The final patient-level diagnosis for both baselines is based on a majority vote aggregation strategy. A patient is classified as depressed if the majority of their individual text segments are predicted as positive by the model.

[1] https://github.com/Tjaardo/SMART-for-Depression-Classification

Basic MentalBERT

The first baseline served as a zero-shot performance assessment of the pre-trained MentalBERT language model from HuggingFace. This base model was loaded with a standard binary classification head but without task-specific fine-tuning. We evaluated the model on the test set immediately after initialisation. Because the weights in the classification head are initially random, this baseline tested whether the pre-trained MentalBERT embeddings already cluster depressive and healthy language far enough apart that even untrained linear mappings can separate them. This set the minimum performance level for our pipeline.

Single-task Fine-tuned MentalBERT

The second baseline evaluated whether the integration of nine expert heads with attention mechanisms and the multi-task learning actually provides clear benefits over standard transformer classification. This model used the standard BERTForSequenceClassification architecture, which directly maps the [CLS] token from the final layer to a binary output. To ensure a fair comparison with our main model, this baseline was trained with a weighted cross-entropy loss to account for the class imbalance in our dataset. This baseline was fine-tuned for three epochs with a learning rate of $2 \cdot e^{-5}$, a weight decay of 0.01, and a batch size of 16.

By comparing our primary model against this single-task fine-tuned baseline, we could directly evaluate the value of our symptom-based approach.

Ablation Study

To evaluate the impact of the binary classification head in the SMART architecture, we conducted an ablation study that trained a variant of the SMART model that removed the binary diagnosis head and consequently the binary cross-entropy loss from the final loss function. This model was therefore only required to predict the eight PHQ-8 severity scores.

To ensure comparability with the main architecture, this model variant was evaluated identically by summing the predicted symptom scores and making a binary decision based on a decision threshold, as explained in Section 4. This experimental design allowed us to determine whether the SMART model's performance is primarily driven by the symptom prediction task or the binary classification task.

Evaluation Metrics

To evaluate the performance of the models for the binary classification (depressed or healthy), we utilised precision, recall, and the macro F1-score to account for the bias in the DAIC-WOZ dataset. For the single-task baselines, the binary diagnosis was determined with a simple majority vote over all segment-level predictions per patient. In contrast, our SMART model's binary diagnosis was calculated by comparing the summed PHQ-8 scores against the calibrated threshold, as detailed in Section 4. Since the baseline models only predict binary labels, a symptom-level evaluation can only be done on the SMART architecture. We assessed the model's ability to predict the specific severity scores of the eight PHQ-8 symptoms using the Mean Absolute Error (MAE) and Mean Squared Error (MSE). MAE provides an intuitive measure of the average error, while MSE strongly penalises larger prediction outliers.

Explainability Setup

We established the theoretical background of our interpretability methods in Section 2. This section focuses on the concrete setup we used in the experiments. All analyses were done on the test set to evaluate the model's ability to generalise.

Correlation Analysis

We verified whether our proposed multi-task architecture learned to isolate specific clinical symptoms by conducting a correlation analysis. During evaluation, we extracted the predictions of the eight symptom heads for all text segments. We then aggregated them by computing the mean score per patient.

A Pearson correlation matrix was then calculated across these aggregated patient-level symptom vectors. To establish a baseline for clinical plausibility, this predicted matrix was compared against the correlation matrix created from the PHQ-8 ground truth scores of the dataset.

Latent Space Analysis

After examining individual tokens and layers, we conducted a latent space analysis to understand the model's global representation of clinical language. Therefore, we extracted the [CLS] token embedding from the final hidden layer of the MentalBERT encoder for each text segment. Although the attention mechanism in our proposed architecture attends to the entire hidden sequence rather than relying just on the [CLS] token, this specific token still holds an internal summary of the input sequence. By investigating how these vector embeddings are positioned relative to each other, we could observe how the model internally structured patient representations.

To achieve this, the 768-dimensional embeddings were averaged per patient to create a patient-level representation. We normalised the embeddings using a StandardScaler before projecting them into a two-dimensional space using UMAP ($n\_neighbors = 10, min\_dist = 0.3$).

We applied this projection across our proposed SMART model and the two baseline models. Thereby, we could directly and visually compare how task-specific fine-tuning clusters patients based on their semantic representations. We visualised each latent space projection using two distinct mappings:

- **Binary Diagnosis:** The first mapping visualised the binary ground truth to separate healthy controls from depressed patients.
- **Continuous Gradient:** The second mapping applied a continuous colour scale to the projections. For our proposed SMART architecture, this gradient represented the predicted aggregated severity scores. For the single-task baselines, which do not predict individual symptom scores, the gradient visualised the prediction probability of the depression diagnosis.

SHAP Feature Attribution

To better understand how our SMART model made its predictions, we used SHAP. Transformer models heavily rely on context [Vaswani et al., 2017], therefore, we split our experiments into a global and local analysis.

Global Feature Importance

The goal of the global analysis was to identify the most important words that led to the predictions for all eight symptom heads. We applied the Shap explainer directly to the raw model logits to keep the SHAP values accurate. As described in Section 2, the WordPiece tokeniser often splits words into smaller pieces (e.g., *introverted* into *intro* and *##verted*). To prevent losing information, we merged these word pieces and their SHAP values back into complete words. Finally, we cleaned the words by converting them to lowercase, removing punctuation and filtering out standard stop words as well as common filler words (e.g., *really*, *just*). During initial tests, we noticed that the results showed a frequency bias. Common words always showed up in the results simply because they appear so often in the dataset. To find the actual important words, we calculated the mean positive impact $M_{w,i}$ for a given word w and a symptom head i. Instead of just summing together all SHAP values, we only added the SHAP values for a word and divided them by the word's total occurrence count C_w. We also have a threshold for the minimum frequency of a word with $C_w \geq 3$:

$$M_{w,i} = \frac{1}{C_w} \sum_{k=1}^{C_w} max(\phi_{w,i}^{(k)}, 0),$$

Where $\phi_{w,i}^{(k)}$ represents the SHAP value of the k-th occurrence of the word. This way, the results only highlighted words that strongly influenced the symptom heads when they appear, rather than words that are just used frequently.

Local Interpretability

While the global analysis showed general importance, it could not capture individual feature importance in a given sentence. Therefore, we also performed a local analysis using heatmaps for SHAP on a sentence level. We visualised the SHAP values for specific input sequences from the test set and evaluated how well the model weights different symptoms within the same context. Furthermore, we used these heatmaps for a qualitative error analysis to understand why the model sometimes makes false predictions.

To further isolate the SHAP importance of a specific token from its surroundings, we conducted a small perturbation test where we only passed the single token through our model and calculated its SHAP values for the eight PHQ-8 symptoms.

Word Impact on PHQ-8 Scores

We conducted a brief statistical analysis on the test set. We compared the mean symptom severity scores of input sequences including and excluding a target word. This allowed us to evaluate whether the feature attribution learned by the model aligned with the actual distribution in the dataset.

Layer-wise Probing

To assess the $\mathcal{V}$-usable information across the different layers in our models, we employed different layer-wise probing experiments. The representations $R^{(l)}$ for layer l are extracted at each of the 13 layers. We employed three different target properties Y for a subset of the test data:

- **Word Surprisal:** We measured the linguistic predictability of words. To generate the labels, a pre-trained GPT2 model was used without fine-tuning. The surprisal was calculated as the mean cross-entropy loss of the GPT2 prediction over the input sequence.

- **First-Person Singular Pronouns:** The density of words associated with a focus on self-reference, such as *I*, *me*, *my*.

- **Absolutist Words:** The frequency of words indicating absolute magnitudes like *always*, *never*, or *completely*. These words are known linguistic indicators of depression.

$\mathcal{V}$-Usable Information

We utilised Ridge Regression with $alpha = 1.0$ to map the layer representations $R^{(l)}$ to the target properties Y. As described in Section 2, we estimated the $\mathcal{V}$-information ($I_{\mathcal{V}}$) by comparing the predictive entropy of the trained probes against a naive null-predictor. This null-predictor always outputs the mean of the target values, with its predictive uncertainty being estimated as the natural logarithm of the MSE. The uncertainty of the trained probes was estimated in the same way. The final usable information gained at layer l was then computed as the difference between these two entropies:

$$I_{\mathcal{V}}(R^l \to Y) = ln(MSE_{null}) - ln(MSE_{probe})$$

We repeated this calculation for all 13 layers across our three architectures (SMART, baselines) to again have a direct comparison and evolution of the feature utilisation.

Chapter 5

Results

This chapter presents the experimental results We begin with listing the binary diagnosis classification performance, including the results of the ablation study in Section 5. Section 5 then shows the model's ability to predict the individual PHQ-8 symptom scores. We look at the correlation between the predicted symptom severity scores in Section 5. Next, Section 5 visually explores the patient representations through a latent space analysis. Finally, Sections 5 and 5 present the findings of our interpretability experiments using SHAP and layer-wise probing.

Diagnosis Classification Performance

The performance of our three models (SMART, Basic MentalBERT, and Single-task Fine-tuned MentalBERT) was evaluated on an unseen test set, which consists of 50 patients. Due to the inherent bias in the DAIC-WOZ dataset, out of the 50 patients, there are 34 healthy controls and 16 depressed patients. The final patient-level diagnoses were aggregated as described in Section 4. We compared the models primarily on their macro F1-score and their recall for the minority class. The results for the three models can be seen in Table 5.1:

Baseline Models

The untrained MentalBERT baseline achieved an overall accuracy of 0.72. The model achieved a macro F1-score of only 0.53 and a recall of 0.12, therefore, it only successfully identified 12% of the depressed patients and classified almost everything as healthy. The single-task fine-tuned MentalBERT baseline showed a similar pattern in classification performance. Even though we fine-tuned with a weighted loss function to mitigate the effects of unequal distribution, the model always predicted the majority class and failed to identify

Table 5.1: Table of the evaluation metrics of the three models (untrained BERT baseline, single-task trained baseline, and SMART). The baseline models achieve a solid accuracy by always predicting the majority label, however, they show a low score for recall and macro F1 for the minority class (depression). Our proposed SMART architecture significantly outperforms the baseline models with an accuracy of 0.80 and a macro F1 score of 0.68. Notably, the expert model scores a recall of 0.81 for the depression class, which is especially important in a clinical domain.

Model	Accuracy	Precision	Recall	Macro F1
Untrained Baseline	0.72	1.00	0.12	0.53
Trained Baseline	0.68	0.00	0.00	0.40
SMART	0.80	0.65	0.81	0.78

a single depressed patient, which can be seen in the recall and precision being 0.00. The resulting accuracy was 0.68 with a macro F1-score of 0.40.

SMART

Our proposed multi-task architecture demonstrated a strong improvement in successfully detecting depressed patients. Our model significantly outperformed the two baselines by achieving an overall accuracy of 0.80 and a macro F1-score of 0.78. Most importantly, the SMART model successfully identified 81% of the depressed patients. This high recall came with a trade-off in precision for the depressed class, which is only 0.65, indicating that the model predicts a higher number of false positives.

The exact number and distribution of true positives, false positives, true negatives, and false negatives for all three models are visualised in the confusion matrices in Figure 5.1

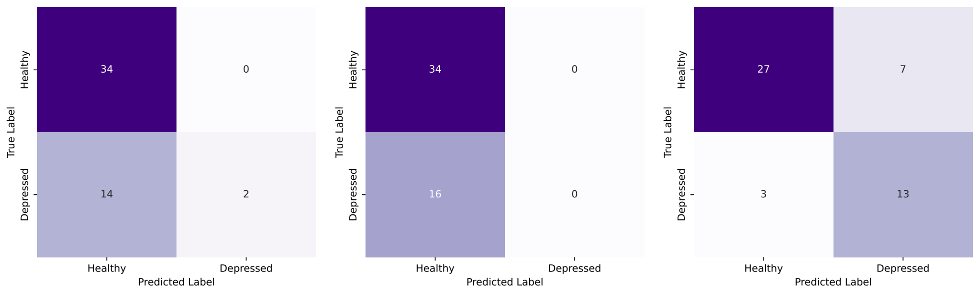

Figure 5.1: Confusion matrices for the untrained baseline (left), trained baseline (middle), and SMART model (right). The baseline models almost always predict the majority class, failing to detect depressed patients. On the contrary, our proposed SMART model successfully identifies most positive cases with only a few false positives and false negatives.

Ablation Study

As described in Section 4, we trained a variant of the SMART model to investigate the impact of the different learning tasks on the model performance. We listed results of this ablation model as well as our proposed SMART architecture in Table 5.2

Table 5.2: Classification performance of the SMART architecture and its ablation variant. The ablation model (trained without the binary diagnosis classification head) still achieves a high macro F1-score of 0.74 and a recall of 0.75 for the depressed class. The SMART model still outperforms the ablation variant with higher scores for every metric.

Model	**Accuracy**	**Precision**	**Recall**	**Macro F1**
Ablation Model	0.76	0.60	0.75	0.74
SMART	0.80	0.65	0.81	0.78

Symptom Score Regression Performance

In addition to the global diagnosis, we evaluated the model's ability to predict the individual severity scores for the eight individual PHQ-8 symptoms. This allowed us to see the performance of each of the eight symptom heads. We evaluated the performance using the MAE and the MSE at the patient level. The results can be seen in Table 5.3.

Table 5.3: Patient-level MAE and MSE across the eight PHQ-8 symptoms for the SMART architecture. The MAE consistently stays below 1 for all symptoms, with the lowest error of 0.621 occurring for Psychomotor Skills and the highest error of 0.956 for Sleep.

Symptom	**MAE**	**MSE**
Interest	0.743	0.927
Mood	0.678	0.703
Sleep	0.956	1.184
Energy	0.868	1.066
Appetite	0.843	1.040
Self-esteem	0.771	0.840
Concentration	0.841	1.063
Psychomotor Skills	0.621	0.879
Mean MAE	0.790	-
Mean MSE	-	0.962

Our proposed architecture achieved a mean MAE of 0.790 and a mean MSE of 0.962 across all symptom scores. We can see from the results that the prediction error varied across the different symptom heads. The lowest MAEs appeared for psychomotor skills (0.621) and Mood (0.678). In contrast, the highest errors were observed for the predicted scores for Sleep (0.956) and Energy (0.868). We visualised the errors between predicted and true symptom scores in Figure 5.2.

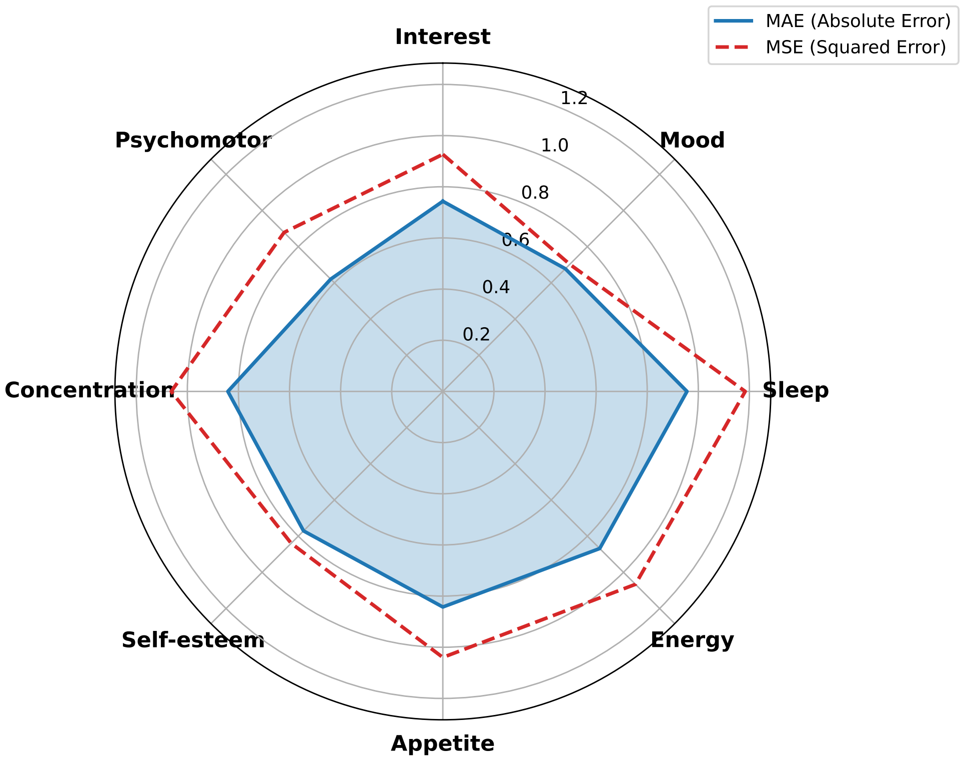

Figure 5.2: Error distribution across the eight symptoms of the PHQ-8 framework. We visualised the patient-level MAE (blue) and MSE (red) for the SMART architecture. The highest error scores can be observed for the Sleep dimension, while Psychomotor skills and Mood show the lowest error scores.

Symptom Score Correlation

The results for the correlation analysis of the true PHQ-8 scores are visualised in Figure 5.4, and the correlations between the predicted scores are shown in Figure 5.3.

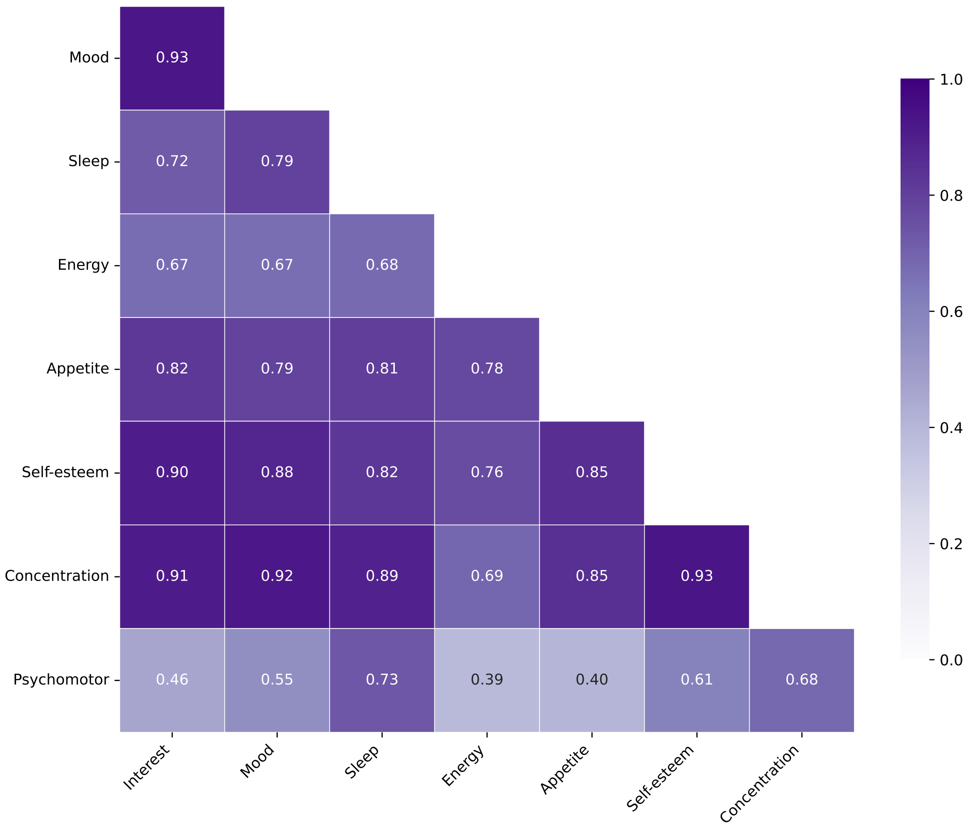

Figure 5.3: Pearson correlation matrix of the PHQ-8 severity scores predicted by the SMART model. The matrix shows the internal degree of correlation learned by the eight distinct symptom heads. There are high positive correlation scores between almost all dimensions. The highest correlation can be observed between Mood and Interest. The lowest correlation was formed between the Psychomotor symptom head and the others, with a minimum of 0.41 between Psychomotor and Sleep.

True Correlations

The true correlation between the severity scores of the PHQ-8 symptoms showed moderate to strong positive correlations across most dimensions, with the highest correlations observed for Mood and Interest (0.76), Sleep and Energy (0.74), as well as Self-esteem and Interest (0.74), and Self-esteem and Mood (0.74). The lowest correlations appeared between the scores for the psychomotor symptom and the others, dropping to 0.41 with Sleep, Appetite, and Self-esteem.

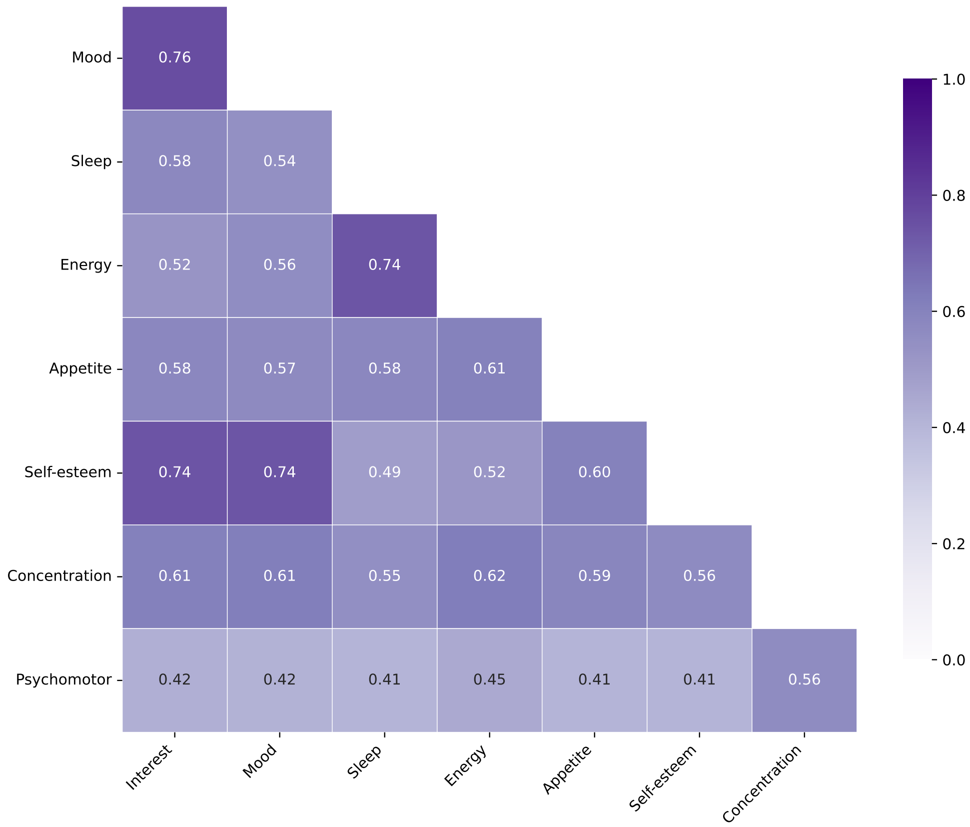

Figure 5.4: True Pearson correlation matrix calculated for the true labels of the DAIC-WOZ dataset. This matrix serves as a baseline used to compare with the proposed model's predictions. We find the highest correlation score between Mood and Interest. The lowest correlation score occurs between the psychomotor dimension and the other 7 symptoms, reaching a minimum of 0.41 with Sleep, Appetite, and Self-esteem.

Predicted Correlations

In the predicted correlation matrix, most symptom scores showed a strong positive relationship, often exceeding the 0.70 threshold. The highest correlations were found between Interest and Mood (0.93), Concentration and Mood (0.92), and Concentration and Interest (0.91). Conversely, some symptoms showed notably lower correlation values. Psychomotor skills had the lowest overall correlation with the other symptom heads, dropping to 0.39 with Energy and 0.40 with Appetite.

Latent Space Analysis

The UMAP projections at the patient level are visualised in Figures 5.5, 5.6, and 5.7, showing the latent space geometry for the untrained baseline, the trained single-task baseline, and the proposed SMART model.

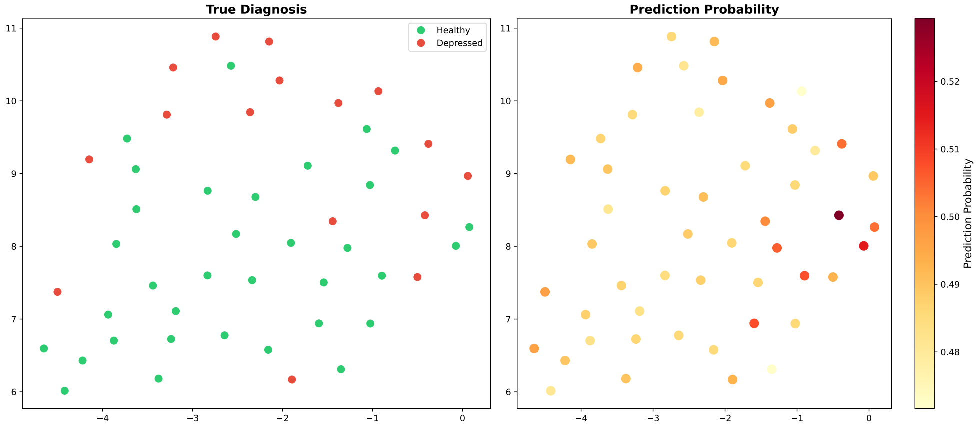

Figure 5.5: UMAP projection of the patient-level [CLS] token embedding for the untrained baseline. The projections are colourised by the binary classification label (left) and the prediction probability for the binary classifications (right). There is no clear clustering or spatial pattern visible, and the embeddings are randomly scattered in the latent space.

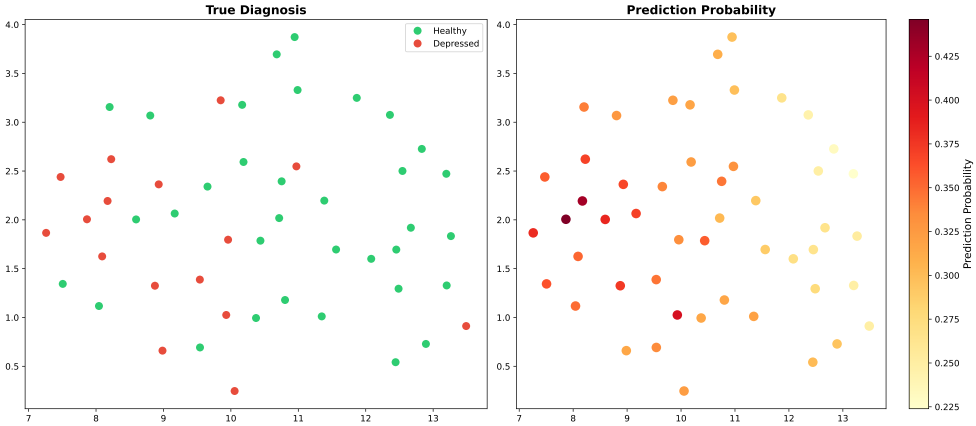

Figure 5.6: UMAP projection of the patient-level [CLS] token embedding for the trained single-task baseline. We colourise the projections by the binary classification label (left) and the prediction probability for the binary classifications (right). We can see a slight ordering, grouping depressed patients on the left and healthy patients on the right.

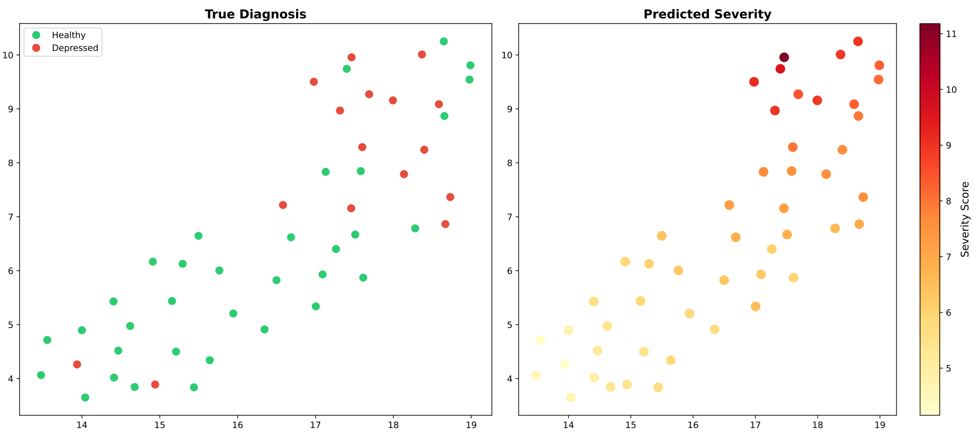

Figure 5.7: UMAP projection of the patient-level [CLS] token embeddings for the SMART architecture. The colours represent the binary classification label (left) and the sum of the predicted severity scores (right). We can see a distinct clustering of the embeddings in depressed and healthy patients. The predicted severity scores lie on a linear axis and show a clear gradient.

Untrained Baseline

In the untrained model, the latent space showed no structural clusterings or patterns regarding the labels. Patients with and without depression were scattered evenly across the space. Consequently, the colour distribution representing depression classification probability appeared random across the entire latent space, and there was no directional gradient observed.

Trained Baseline

The trained single-task baseline model showed only small changes in spatial alignments. The visualisation of the classification probabilities revealed the beginnings of a slight directional gradient, but the binary diagnosis labels still overlapped significantly, indicating a lack of spatial separability between the two classes.

SMART

In contrast to the baselines, our proposed multi-task architecture revealed a distinct clustering and spatial pattern. The projection showed a visible separation between healthy controls and depressed patients, grouping them in distinct clusters. Furthermore, instead of visualising the classification probabilities, we plotted the summed severity scores of the PHQ-8 symptoms. The results exhibited a clear, continuous gradient across the latent space. The representations transitioned from low severity scores in the lower-left to high severity scores in the upper right.

SHAP Interpretability

The global feature importance, calculated with the mean positive SHAP values across the entire test set, is visualised in Figure 5.8.

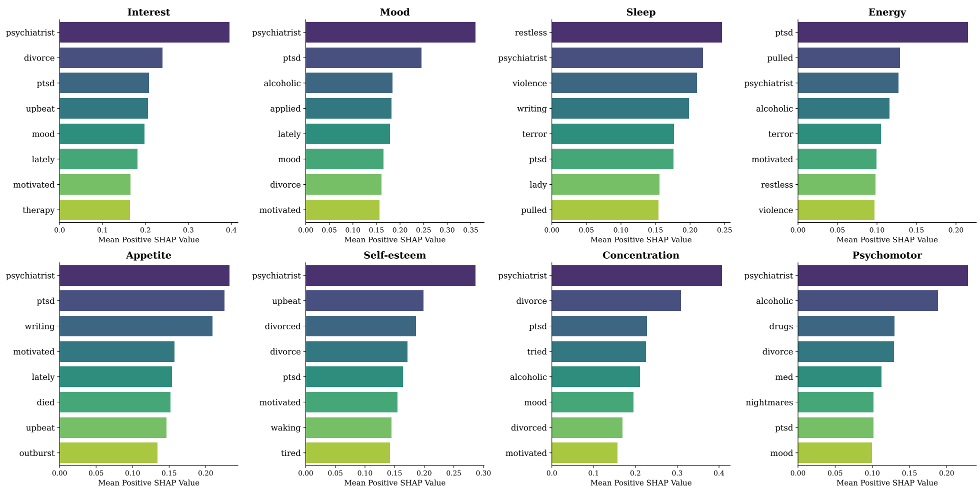

Figure 5.8: Mean global SHAP feature attribution across the eight PHQ-8 symptoms. We identify clinical anchor terms like *psychiatrist* or *ptsd*, as well as symptom-specific words like *restless* and *terror* for the Sleep head.

The results demonstrated a significant degree of lexical overlap among the top important words for the eight symptom heads. Clinical or trauma-related words such as *psychiatrist*, *ptsd*, and *alcoholic* consistently appeared as most important across multiple distinct symptom heads. Despite this global overlap, we could find symptom-specific vocabulary in the results. For instance, the words *restless* and *terror* scored a high rank, exclusively for the Sleep symptom, while *drugs* and *med* appeared within the Psychomotor dimension. Furthermore, the global analysis highlighted words that seem counterintuitive, such as *motivated* or *upbeat*, that also produced positive SHAP values, thereby actively increasing the predicted depressive severity scores.

Local Sentence Analysis

To observe the contextual behaviour of the multi-task attention heads and investigate any unintuitive patterns found in the global analysis, we generated section-level SHAP heatmaps.

Figure 5.9 visualises the importance of words for the input sequence "*Very easy, I just go lay down and listen to some music for a little while and then I fall asleep*" from the test set. The results showed that the word *easy* received a strong negative SHAP value of down to -0.76 across the regression heads, excluding the psychomotor symptom, lowering the predicted severity scores. The tokens *fall* and *asleep* showed values closer to zero.

Figure 5.9: Local SHAP attribution heatmap for a sleep-related input. There are strong negative attributions for the word *easy*(-0.76), while sleep related words like *fall* and *asleep* show scores close to zero.

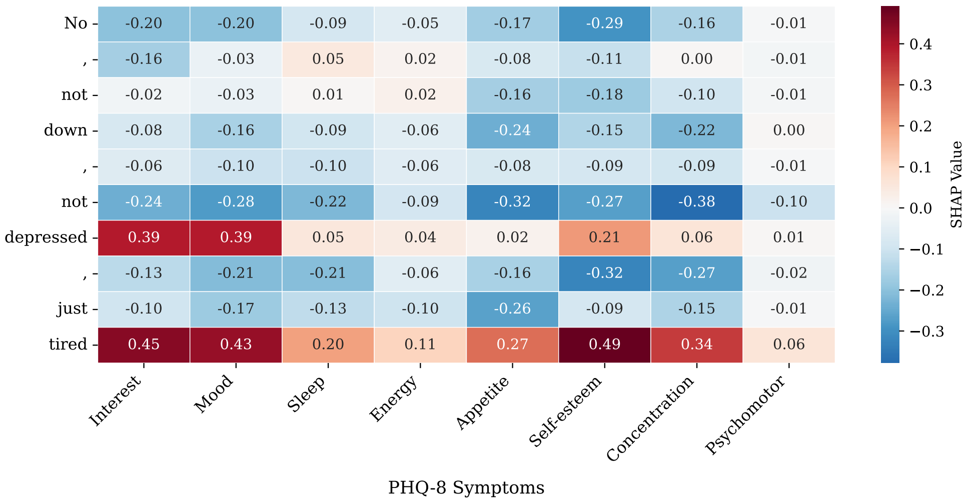

Figure 5.10: Local SHAP attribution heatmap for a negated sentence. The negation word *not* has a negative impact that is counterbalanced by a positive attribution of the tokens *depressed* and *tired*.

Figure 5.10 shows the SHAP attributions for the sentence "*No, not down, not depressed, just tired.*" The negation token *not* that occurred before *tired* received a negative score of down to -0.38. The positive values of the words *depressed* and especially *tired* were significantly higher than those of any other word. The highest importance of the token *tired* appeared for the Self-esteem symptom head (0.49), while the value for the Energy symptom was comparatively low (0.11).

To further isolate the impact of this specific token from its surrounding context, we conducted a perturbation test by providing the model with the isolated word *tired* and calculated the SHAP values.

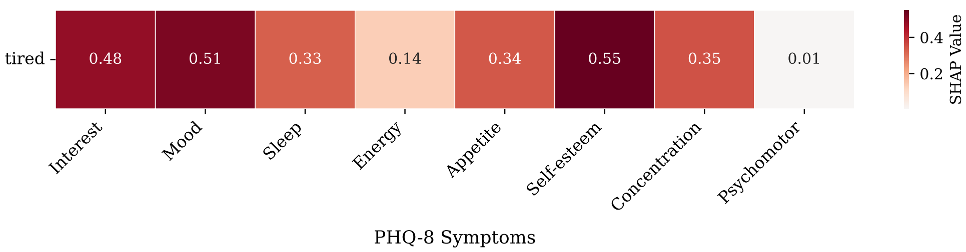

Figure 5.11: SHAP attribution heatmap for the isolated word *tired*. The heatmap shows a positive score for the Self-esteem dimension of 0.55 and a low attribution for Energy with 0.14.

As shown in Figure 5.11, the isolated token produced even higher SHAP values for almost all of the Symptoms compared to the contextualised sentence.

Again, the highest importance of *tired* appeared for Self-esteem (0.55) and the lowest for Psychomotor (0.01) and Energy (0.14).

Finally, Figure 5.12 demonstrates the feature importance for the sentence "*Um, I have a friend named ... He's pretty inspirational. He keeps me uh, motivated as far as my higher level thinking goes.*" Most notably, the model assigned positive SHAP values to the word *motivated* across multiple symptom heads, with the highest value for Self-esteem (0.34) and Mood(0.28).

Impact of *tired* on True PHQ-8 Severity Scores

We calculated the true PHQ-8 score distribution for all test-set segments containing the word *tired* and compared them to the segments without this word.

As visualised in Figure 5.13, the mean true severity scores were consistently higher across all eight PHQ-8 symptoms when the word *tired* was present in the text segment. The largest increase occurred in the Energy dimension, which increased from 1.16 to 1.88. On the other hand, the smallest difference between including and excluding this token was recorded for Self-esteem which went from 0.97 to 1.12. The absolute highest mean values were recorded for Energy and Sleep, and the lowest mean values appeared for Psychomotor and Self-esteem.

Layer-wise Probing and $\mathcal{V}$-usable Information

As described in previous chapters, we wanted to quantify how cognitive and psycholinguistic features are encoded across the individual layers of our networks by computing the $\mathcal{V}$-valuable information at each layer. We visualised the information for cognitive surprisal, self-reference, and absolutist words in Figure 5.14.

Cognitive Surprisal

In terms of cognitive surprisal, the $\mathcal{V}$-usable information started at a low value at the initial embedding layer for all models (≈ 0.95). As the data passed through the network, the usable information monotonically increased. Our proposed SMART architecture consistently demonstrated a slightly higher extraction of usable information across the early and middle layers compared to the two baseline models. All models reached their peak in information gain between layers 9 and 11 (≈ 2.75), indicating that this was the point where structural context was most accessible.

First-Person Singular Pronouns

The probing results for the self-reference, measured by first-person singular pronouns, showed a steep initial increase directly after the embedding layer for

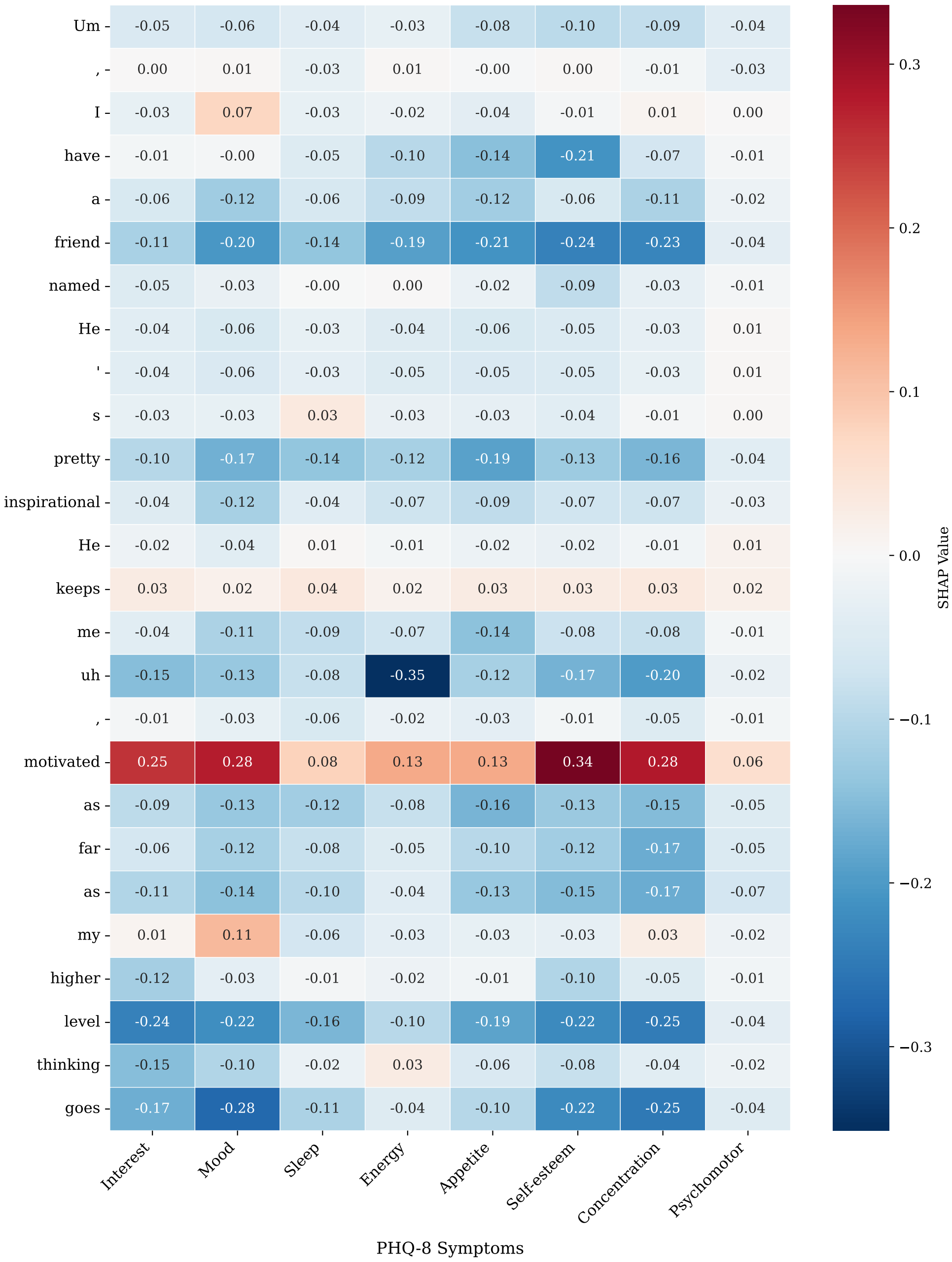

Figure 5.12: Local SHAP attribution heatmap for a positive sentiment sequence. The model assigns high SHAP values to the usually positive word *motivated*, with the highest value occurring for Self-esteem with 0.34.

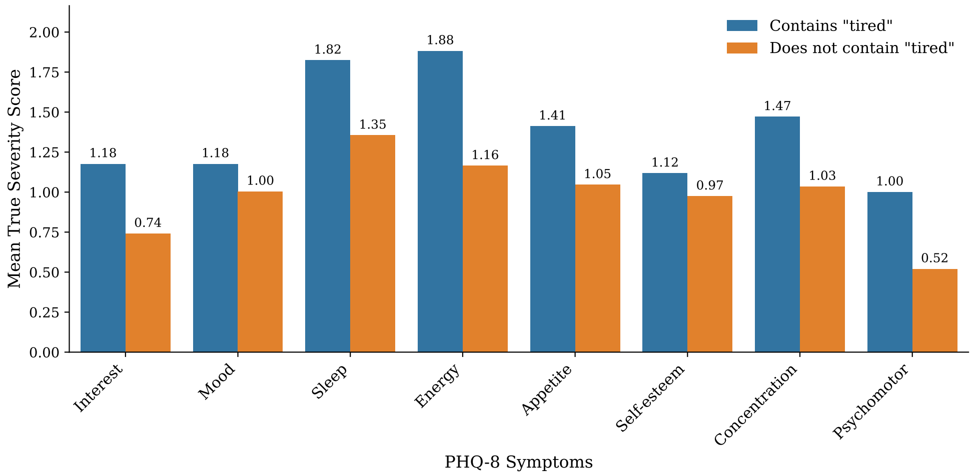

Figure 5.13: True PHQ-8 score distribution with and without *tired*. If the word *tired* occurs in the input sequence, the severity scores for all symptoms increase. The highest difference appears for Energy, which rises from 1.16 to 1.88.

all models. The SMART architecture encoded more usable information in the early layers (1 to 3) compared to the baselines. However, from layer 3 onwards, the amount of accessible $\mathcal{V}$-usable information diverged. The baseline models maintained a high level of information gain until the final layers, peaking between layers 3 and 5, while the SMART model showed a steady decline for all following layers.

Absolutist Words

The probing for absolutist words again revealed differences between the SMART architecture and the baseline models. While the SMART model outperformed the baselines in the information gain in the lower layers (1 to 5), the baselines encode more $\mathcal{V}$-usable information from layer 5 onwards. The SMART architecture showed a significant drop in information gain after layer 9, whereas the baseline model peaked at layer 10.

General Difference in Later Layers

Across all target properties, we observed that while all models gained information in the early and sometimes middle layers of the network, they differed in the final layers. The baseline models tended to maintain higher $\mathcal{V}$-usable information until the end, however, the SMART model consistently showed a drop in the information gain in later layers. This difference will be evaluated in the following chapter.

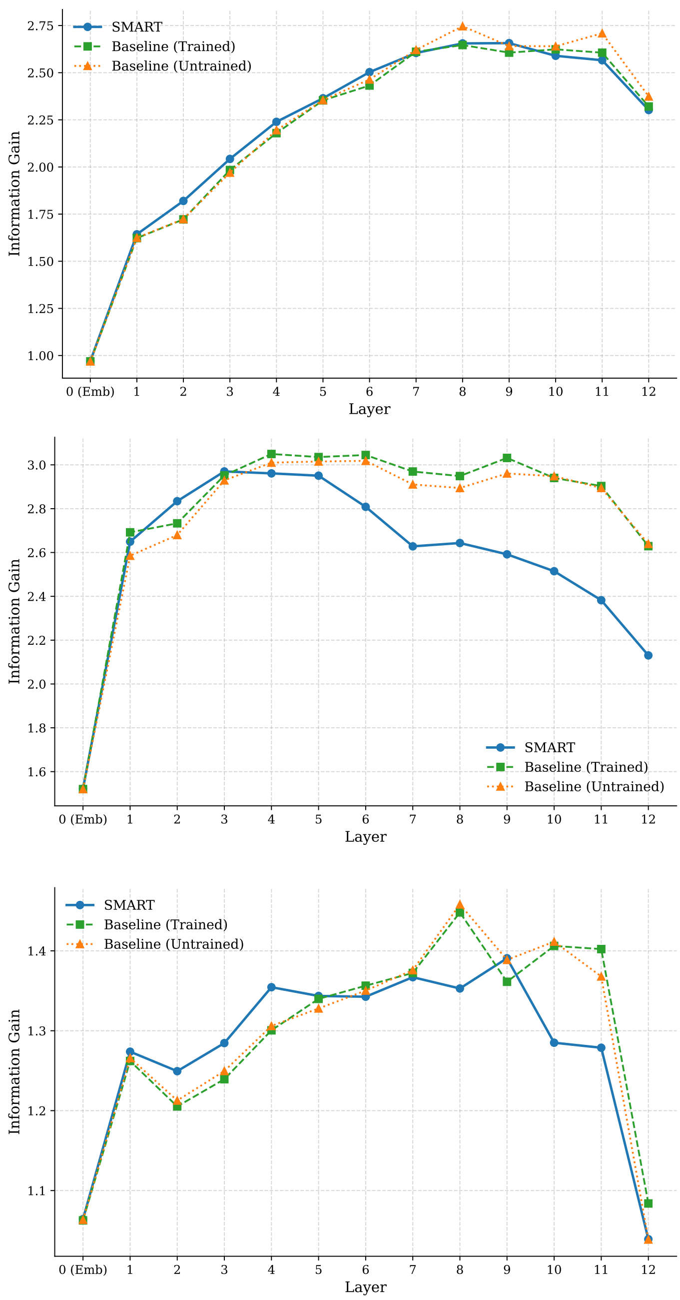

Figure 5.14: Layer-wise probing of $\mathcal{V}$-usable information across the network layers for the SMART model and the two baselines. The investigated target properties of the probing are cognitive surprisal (Top), self-reference (Middle), and absolutist words (Bottom). While all models show an increase in information gain in the early layers for all target properties, the SMART architecture starts to decrease significantly earlier and more strongly than the two baseline models. This behaviour is mostly visible when probing for self-reference (Middle).

Chapter 6

Discussion

A main goal of this thesis is to overcome the limitations and challenges of binary depression classification by proposing a symptom-based multi-task learning architecture, called SMART. We integrated a symptom-specific attention mechanism that should allow the model to learn two tasks simultaneously, namely predicting the eight severity scores for the symptoms of the PHQ-8 framework together with the binary diagnosis. The results shown in Chapter 5 provide promising evidence that our proposed architecture not only improves model performance but also helps the model to structure the language internally in a more clinically grounded way.

This chapter interprets our findings and evaluates the full capabilities and internal mechanisms of our proposed model architecture. Following a summary of the main findings in Section 6, Section 6 discusses how the multi-task objective can act as a structural regulariser to overcome feature collapse. Section 6 then analyses the variance in predictability between cognitive and somatic symptoms, while Section 6 investigates the correlations between the predicted symptom scores. We explore the model's internal representations in Section 6, where we look at the latent space projections of the model's [CLS] token embeddings. Finally, Section 6 evaluates the clinical logic of our SMART model using SHAP, and Section 6 discusses the layer-wise probing results.

Summary of the Main Findings

The empirical evaluation of the SMART architecture hinted at several key findings regarding the capabilities of multi-task symptom learning in depression classification and clinical language processing in general. First, the results demonstrated that our proposed architecture successfully prevents the feature collapse that caused the trained and untrained single-task baseline models to fail. By forcing the network to simultaneously optimise for the eight PHQ-8 symptom heads, together with a correlation penalty, the network is forced to recognise distinct clinical dimensions of depression in the language.

Furthermore, an ablation study confirmed that relying entirely on symptom regression for the final diagnosis is highly effective. This proves that, by forcing the network to focus on individual symptoms, it builds a clinically grounded internal representation that successfully detects depression without needing a direct binary classification target. Secondly, we found that the predictability varies significantly between cognitive and somatic symptoms and highlights a limitation of text-only models. Emotional factors such as depressed mood can be detected from text much more easily than factual symptoms like energy or Psychomotor skills that often manifest in acoustic and visual signals. Despite these limitations, our proposed model successfully captured the relative correlations between the individual symptoms without collapsing into identical outputs. This clinical understanding was structurally proven by the latent space projections, where the patient embeddings formed a continuous severity gradient transitioning from healthy controls to severe cases of depression.

Finally, layer-wise probing and SHAP interpretability analysis provided deeper insights into the internal logic of the network. The probing experiments showed that while all models encoded cognitive proxies like self-reference and absolutist words in their early layers, the multi-task learning objective forced the SMART architecture to abstract from these features in its deeper layers. The model prevented shortcut learning and feature collapse by transitioning from simple heuristics to high-level clinical representations. On a token level, SHAP visualised that the attention heads can use context to adjust the symptom scores, but the model still struggles with negations. Additionally, we found a tendency towards semantic over-generalisation, where tokens like *tired* are just used as global proxies for depression without understanding their underlying clinical meaning.

Overcoming Feature Collapse with Multi-task Learning

One of the contributions of this thesis is to demonstrate how single-task models suffer from representation degeneration in clinical domains and how our proposed multi-task architecture solves this problem. Traditionally, binary classifiers such as the evaluated baseline models are trained to predict a final binary label, minimising the cross-entropy loss.

The Problem with Single-task Objectives

As we showed in Section 5, the trained single-task baseline completely failed to identify depressed patients by always predicting the majority label and therefore achieving a recall of 0. It can be argued that this is purely due to class imbalance, however, the latent space projections in Section 5 demonstrated that this happened due to feature collapse. The baseline models failed to form a meaningful structural separation. The single-task models found an exploit to reduce their loss. Instead of learning the digital biomarkers of depression in the language, the network collapsed its internal representations into a single cluster that satisfied the loss function for the majority class, ignoring anything else.

Symptom-level Predictions as a Regulariser

Our SMART architecture prevents the feature collapse by being trained on a multi-task objective, namely, predicting the severity scores from the PHQ-8 framework. By forcing the model to simultaneously optimise the distinct symptom heads, the model is unable to collapse into a single heuristic. If one of the heads started predicting a generic depression signal, the specific loss from this head would penalise the network.

Additionally, we introduced the correlation penalty to the final loss function to ensure the different dimensions remain separate. This penalty actively penalises the network if the outputs of the eight symptom heads become highly correlated and not orthogonal in the prediction space. It guarantees that the network does not simply copy a global prediction signal across all symptom heads but instead learns distinct features for every head.

The results from Section 5 directly show that the network is forced to recognise distinct clinical dimensions of depression in the language.

Findings from the Ablation Study

Our ablation study comparing the SMART model with a variant that removed the binary classification head and predicted only symptom scores provides evidence that the linguistic understanding of the SMART architecture is driven almost entirely by the continuous symptom regression rather than by the explicit binary classification objective. The ablation model still achieved a good Macro F1 score of 0.74 and successfully identified most of the depressed patients. This can be explained by looking directly at the dataset itself. The binary diagnosis is not an independent variable but rather a linear combination of the symptom severity scores. The ablation study shows that by forcing the model to only focus on the symptom scores, it can approximate a binary threshold without requiring an explicit binary classification target.

Furthermore, this finding is favoured by the weighting applied in the final loss function of the SMART architecture. By weighting the MSE of the symptom prediction significantly higher (0.7) than the binary cross-entropy loss of the diagnosis (0.3), the network is already biased towards prioritising the symptom extraction.

Since our pipeline builds the final diagnosis directly from the aggregated symptom scores, the strong performance of the ablation model confirms that this symptom-learning approach is highly effective. Instead of allowing the network to search for a generic global pattern, which led to feature collapse in the baselines, the multi-task symptom regression acts as a structural regularizer that preserves the psycholinguistic information in the input, enabling a much more clinically grounded internal representation.

Predictability Varies Across Cognitive and Somatic Symptoms

The global binary classification performance proves the overall accuracy of the SMART model, however, we also evaluated the MAE and MSE of each symptom head to gain a more detailed insight into the model performance. Our model achieved a mean MAE of 0.790 across all dimensions, but showed variance in the predictability of cognitive and somatic symptoms.

High Predictability in Cognitive States and Data Imbalance

The lowest prediction error was recorded in the Mood and Psychomotor dimensions. For Mood, this linguistically makes sense as it is a continuous state that is evident in negative sentiment, absolutist words, or emotional phrasing throughout the interview. The attention mechanism, therefore, has many signals that help make accurate predictions. On the other hand, the low prediction error for psychomotor severity scores is most likely not due to our model architecture but rather the data distribution in the dataset. Most patients show low severity scores (see Table 4.2), therefore, the model gets away with always predicting a low score.

Signal Sparsity and Modality Gap

The highest prediction error appears for Sleep and Energy symptoms. The difficulty in accurately predicting these severity scores highlights major limitations of a text-based model.

Unlike mood, sleep is an episodic, factual topic. In the DAIC-WOZ interviews, sleep quality is typically only discussed briefly. If this short section is vague and not detailed, the transcript lacks clear textual signals needed for accurate predictions. We can call this signal sparsity.

Secondly, there is a modality gap. Energy levels are primarily communicated through multimodal physical outputs like acoustic (e.g., low volume, slow speech) and visual features (e.g., lack of facial expressions, posture). The transcripts from the DAIC-WOZ interview completely remove these physiological signals. A patient might say that he is doing okay, but sounds completely exhausted. Therefore, unimodal models such as SMART struggle to accurately predict the physical energy dimensions.

Error Analysis

We wanted to better understand the high error score in the Sleep or Energy dimension, therefore, we looked at the data that showed high error scores. A prime example is patient 696, who has a significant discrepancy between the true Sleep severity score (3) and the model's mean prediction (0.98). We reviewed this patient's interview and found clear signals regarding sleep problems that, however, were highly localised. In the fifth text segment, the patient states: "*not too easy, I stay up a lot. I often wake up when they're already asleep.*" Despite this clear evidence, the predicted score remained low. This is most likely due to the signal sparsity combined with our decision to perform a segment-level mean aggregation.

While continuous symptoms are visible throughout the entire interview, factual symptoms are only discussed in certain text segments of the interview. Even when the model successfully predicts a high score for these text segments, when calculating the mean score for the final output, this high score will be suppressed.

Another example of a high prediction error for Sleep is patient 637. This patient gave a Sleep symptom score of 0, however, the model predicted a mean severity score of 1.62. Looking at the interview segments, we found that the patient said: "*I am just tired, but pretty fine energy-wise*" or "*I have a lot of lethargy*", despite reporting a 0 in the PHQ-8 questionnaire for the Sleep symptom. The model correctly identified the linguistic biomarkers of the symptom, but was penalised for the true severity score.

This hints at the possibility of self-reporting bias in the dataset. Patients sometimes provide inaccurate or inconsistent answers on questionnaires, however, during a conversational interview, especially with a virtual agent, they may verbalise their conditions more accurately and openly [Rosenman et al., 2011, Lucas et al., 2014]. This is proof that some of the model's prediction errors are not due to the architecture or training but rather due to inconsistent data.

Correlation of Symptom Severity Scores

We calculated the correlations between the true severity scores as well as the predicted severity scores to investigate whether the model learned the correct relationships between the different symptoms. Depression is not an isolated disorder but a combination of different symptoms that appear together [Fried and Nesse, 2015]. By comparing the correlation matrices of the true data with the model predictions, we can check if the model has learned this clinical fact.

SMART successfully learns Symptom Score Correlations

The true severity scores of the DAIC-WOZ dataset show that depression symptoms are highly interconnected, e.g., there is a strong positive correlation between symptoms like Mood and Interest. When investigating the correlation matrix of the model's predictions, we can see that it closely mirrors this ground truth. The model successfully recognises that if a patient talks about having a depressed mood, they are also likely to suffer from a loss of interest. This shows that the eight regression heads learned clinically grounded relationships rather than making isolated guesses.

Preserved Symptom Independence

A risk with multi-task learning is that the different prediction heads collapse and learn to output the same value for most tasks. If our proposed architecture had done this, the predicted symptom scores would show correlations close to 1. Our results clearly show that this did not happen. The model captured the natural variance of the symptoms, which is most visible for the psychomotor symptom. Both in the true scores and the model's predictions, this symptom dimension showed a significantly lower correlation with the other scores. This also makes sense from a clinical perspective. Psychomotor skills are a specific physical symptom that does not always correlate with emotional symptoms like Self-esteem or Interest. Our results are strong evidence that the model's eight regression heads remained independent. Each symptom head focuses on its distinct feature and psycholinguistic targets.

Predicted Correlations are significantly higher than the Ground Truth

Even though our proposed model architecture is able to learn the correct relative relationships between the symptoms, the absolute correlation values in the predictions are generally higher than the ground truth. This happens most likely due to the different symptom heads being fed with the token embeddings that contain contextual information. Each token already encodes some semantic information from the surrounding sequence. Consequently, even though the distinct symptom heads focus on different markers, the underlying semantic space has some overlap, resulting in higher correlation scores.

Latent Space Topology

We hypothesised that multi-task symptom learning enables a deeper understanding of clinical language by separating the distinct dimensions of depression. Although our classification metrics confirm that our model is accurate, to investigate the model's internal representations, we need to examine the relationships between the predicted symptoms and the patterns in the latent space.

Binary Cluster and Severity Gradient

We can see the impact of the multi-task learning directly in the UMAP projections of the patient-level [CLS] token embeddings in Section 5. As expected, the untrained baseline demonstrated a completely unordered latent space, as the raw MentalBERT embeddings have no understanding of clinical psychology that goes beyond the information gained during pretraining. However, even after fine-tuning, the trained single-task baseline still had a significant spatial overlap between the classes. Because this trained baseline was forced to minimise the binary cross-entropy loss, it failed to capture the slight variations between patients. In contrast, our proposed SMART architecture achieved a clear separation between healthy controls and depressed patients. The true benefit of this model, however, can be evaluated when looking at the latent space colorised by the predicted severity scores (see Figure 5.7). Rather than forming two isolated clusters, which is expected of a binary classifier, the multi-task representations are arranged along a clear, continuous gradient. The embeddings transition from low to high severity scores.

This demonstrates that the model organises patient representations along a structural axis of depression severity. By training the model to predict individual symptom scores, the latent space is stretched, and patients are projected realistically based on their severity.

Emergent Structure in the [CLS] Token

The UMAP projections reveal an interesting property of our proposed architecture. We generated the projections using the patient-level [CLS] token embeddings. In traditional BERT classification models, the [CLS] token is directly connected to a final classification layer and therefore learns all the required information for the final model output during training.

However, in the SMART model, the final diagnosis is not based on the [CLS] token. Instead, the architecture aggregates the information across all individual token embeddings in the input sequence to predict symptom scores. Consequently, the [CLS] token was never explicitly trained to classify the patients. The fact that it still learns and encodes a clear severity gradient demonstrates that the multi-task learning objective successfully propagated the gradients back through the entire network. The model's entire understanding of clinical language was changed and incorporated into the foundational representations.

Evaluating the Clinical Logic with Feature Attribution

While the quantitative metrics, latent space projections, and layer-wise probing results confirm the predictive power of our proposed SMART architecture, we want to further understand how the model extracts and processes psycholinguistic information in the eight distinct symptom heads. The results from our SHAP experiments (see Figure 5) revealed insights into the model's reliance on global clinical markers, contextual separation, and challenges of linguistic structure.

Global Anchor words

The global SHAP analysis showed that the eight symptom heads have a high overlap in words that achieve a high SHAP importance. Rather than entirely distinct vocabularies for each symptom, the model relied on shared clinical and trauma-related tokens such as *psychiatrist*, *ptsd*, or *alcoholic*.

While this lexical overlap might initially appear as a failure of the multi-task and symptom-specific architecture that is trained to learn distinct linguistic patterns, it actually highlights a typical behaviour with shared encoders. Depression is a highly interconnected disorder. Psychological triggers do not just affect one symptom alone. Certain words increase the severity scores for all symptom heads at once. This demonstrates that the model reflects the impact of severe events on Mood, Sleep, Concentration, and Energy levels simultaneously.

Positive Tokens in Depressive Contexts

Furthermore, the global feature importance revealed an unintuitive finding. Positive words such as *motivated* or *upbeat* showed a strong positive impact on the depressive severity scores. One might argue that this is just due to tokenisation or aggregation that could happen, for example, when the model strips away negation words (e.g., *not* from *not motivated*). However, our local analysis (see Figure 5.12) proved that the model assigned high severity logits to the word *motivated* even in contextually positive statements, such as describing a friend who "*keeps me motivated.*"

Psychologically, this makes a lot of sense. The SMART model learned that discussing the active effort to stay motivated often implies an underlying lack of motivation. Healthy participants rarely mention their own motivation in such a detailed way. A high SHAP value score can hint at the model identifying coping strategies and self-reflection. It successfully learned that patients who explicitly speak about their struggle to maintain a positive mental state usually have higher severity scores in dimensions like Self-esteem and Mood.

Contextual Understanding and Negation Errors

The local heatmaps show why just investigating the global SHAP values is not enough. While the global analysis highlighted certain anchor words, the local heatmaps proved that the attention heads can separate information based on context. As seen in Figure 5.9, the model correctly identified the word *easy* as a sign of healthy sleep. It lowered the severity scores and correctly ignored the usual trigger word *asleep*.

However, the local error analysis also demonstrated a structural limitation of the proposed architecture. When a patient negates a symptom (e.g., "*not depressed, just tired.*"), SHAP showed that the model correctly assigned negative values to the word *not*, but the values for the word *depressed* were stronger, which led to a wrong impact on the severity scores. In this specific example, the word *tired* mostly influenced the symptom head responsible for Self-esteem (0.49) instead of the one for Energy (0.11) or Sleep (0.20). To investigate whether this clinical mismatch was caused by a bias in the dataset, we calculated the true PHQ-8 score distribution for all test-set segments containing the word *tired*. We found that in the DAIC-WOZ dataset, the use of the word *tired* has the strongest positive impact on the Energy severity scores and the lowest impact on Self-esteem (see Figure 5.13). Therefore, the SMART architecture directly contradicts the actual data distribution.

Initially, we thought that this could be due to the surrounding negative context influencing the model, but after removing the context and investigating the SHAP values for *tired* alone, we found that the word still had the most influence on the symptom head responsible for Self-esteem, while the importance for Energy or Sleep remained low (see Figure 5.11).

This finding reveals another significant limitation of fine-tuning language models on clinical datasets. Since it is highly common for depressed patients to struggle with tiredness, the model likely did not learn the correct clinical meaning of the word *tired* but rather used it as a proxy for depression. Consequently, whenever the word appears, it acts as a strong trigger for some symptom heads, regardless of the context. This shows that despite our symptom-specific attention mechanism, the model sometimes struggles with clinical logic.

Multi-task Learning Improves Feature Extraction Across Network Layers

We wanted to evaluate the layer-wise probing results from Section 5 to understand how the text is processed and encoded across the hidden layers of the SMART architecture. Specifically, we wanted to understand how the predictive information propagates through the individual layers.

In transformer models like BERT, where language is processed sequentially, early layers typically handle basic syntax, while deeper layers encode contextualised semantic meaning [Tenney et al., 2019]. Layer-wise probing allows us to understand when the model understands the task. If a network retains high $\mathcal{V}$-usable information for simple word frequencies in the very last layers, it strongly suggests that the model relies on some heuristic at the output [Vandermeerschen and De Lhoneux, 2025]. A decrease in the predictability in the final layers or an increase across the middle layers, however, indicates that the network transitions from processing low-level features to building a deep understanding of the input texts [Liu et al., 2019].

By evaluating the specific probing tasks, we can investigate how our proposed SMART architecture prevents shortcut learning and adapts its linguistic understanding.

Probing for Surprisal

The probing for surprisal, which measures the predictability of language and therefore its structural complexity, revealed that all models followed a similar behaviour. They encoded the highest amount of $\mathcal{V}$-usable information in their middle layers. However, the SMART architecture captured slightly more information across the early and middle layers compared to the two baselines. This aligns well with the multi-task objective. By training the model to predict severity scores, it cannot rely on vocabulary alone, especially for psychomotor symptoms that often manifest structurally in language [Buyukdura et al., 2011, Stade et al., 2023]. The model, therefore, internally encoded the structural complexity of the patient's speech.

The binary baselines, on the other hand, focused more on specific trigger words such as *depressed* or *sad* to make predictions and therefore required less structural information.

Probing for Self-Reference

Our results for the layer-wise probing for first-person singular pronouns provided insights into how the models process negative self-reference. The two baselines again showed almost identical trends in the amount of encoded $\mathcal{V}$-usable information. They used self-reference primarily in the later layers close to the final binary classification. This hints at these single-task models learning that a high frequency of first-person singular pronouns correlated with a depressed patient.

The SMART model, on the other hand, processed these pronouns with a completely different trajectory. It encoded this information earlier (layers 1 to 5) but showed a significant decrease in $\mathcal{V}$-usable information in the deeper layers (layers 9 to 12). A plausible explanation for this trend could be the need for contextualisation. The model processed the first-person pronouns early on but shifted its focus in the later layers to the surrounding context to better evaluate what the patient is saying about themselves. Once this was done, the token frequencies were no longer accessible to our probes, proving that the model successfully abstracted these linguistic markers into a clinical understanding.

Probing for Absolutist Words

A similar abstraction process occurred with absolutist words. For a binary classifier, absolutist words are just additional negative tokens. The binary target does not change whether a patient says "*I feel bad*" or "*I always feel completely bad*." Consequently, the baselines retained this lexical information late into the network, indicating that it was used as a simple heuristic for the final diagnosis prediction.

On the contrary, in multi-task symptom learning, the model must predict different severity scores. Absolutist words are linguistic expressions that can be used to quantify the intensity of a symptom or struggle. To distinguish between a low and a high severity score, the network needs to understand the intensity of a statement. Figure 5.14 indicated that our proposed architecture encoded this requirement in the early and middle layers to assess statement intensity. However, in the final layers, $\mathcal{V}$-usable information for these specific words dropped significantly. Instead of using word count as a simple heuristic for the regression task, the network abstracted and translated the linguistic intensity into severity gradients to ensure accurate predictions.

Chapter 7

Limitations & Future Work

This chapter provides a detailed evaluation of the limitations in our architecture, the training process, and the data in Section 7. Building upon these limitations, Section 7 gives an outlook on possible directions for future work in this field of research.

Limitations

Unimodal Constraints and the Modality Gap

We identified a primary limitation of the SMART model in its reliance on textual data. The error analysis across the eight PHQ-8 dimensions revealed a significant variance in predictability between cognitive symptoms and somatic or physical symptoms. Cognitive and emotional states often manifest in vocabulary, syntax, or sentiment, making them more easily accessible to language models like MentalBERT.

In contrast, somatic symptoms are usually found in physical signals. Changes in the Energy levels or Psychomotor skills show primarily in acoustic features such as voice modulation, speech rate, or volume drops [Cummins et al., 2015], as well as visual behaviours like flat affect, reduced gaze fixation, or change in posture [Canales et al., 2010, Girard et al., 2014]. By processing only transcripts of spoken language in the DAIC-WOZ dataset, the model struggles with a modality gap. It has to approximate physical conditions through semantic proxies. When a patient describes their condition using neutral words but speaks in a completely exhausted tone, a unimodal model is blind to these signals in the other modalities.

Mean Aggregation Can Lead to Signal Dilution

The choice to perform segment-level mean aggregation to compute the final patient-level prediction introduces a major structural limitation. Clinical dialogues contain two distinct types of language patterns. These are emotional states and localised factual reports. Symptoms like depressed mood or low self-esteem manifest globally and can be visible in almost every sentence throughout the whole interview. Factual symptoms such as sleep disturbance or changes in appetite are typically only discussed within a small portion of the interview.

Furthermore, patients vary a lot in the amount of spoken language. This creates an unequal number of text segments per patient, which leads to critical signal dilution. If a highly verbal patient generates 60 text segments in one interview but talks about sleep problems in only one specific segment, the SMART model might successfully predict a maximum severity score for that single segment. However, during the final patient-level score aggregation, this local high score is divided by the total number of segments and, therefore, possibly completely suppressed by the neutral Sleep severity scores of the other 59 segments. The current aggregation mechanism lacks the ability to prioritise short, relevant segments over general conversational filler text.

The Model has to Deal With a Lack of Interviewer Context

Our architecture completely excludes the dialogue parts spoken by the virtual interviewer, processing only the isolated text segments of the patients. This potentially leaves out critical contextual dependencies. In NLP, short responses by the patient like "*Yes, sometimes*", or "*No, only at night*" have no semantic value when processed alone. Without knowing the exact question the interviewer asked, the model's attention heads are forced to evaluate the isolated patient responses. The prior example of the sentence "*No, only at night*" could refer to a lack of energy, sleep problems, or sudden panic attacks. By leaving out the corresponding interviewer question, the model loses the conversational context to deal with semantic ambiguities, which could directly lead to false predictions for the individual symptom heads.

Even though this is a topic that is still being debated, some prior research has shown that including the interviewer's utterances in the model training can significantly improve prediction performance [Dai et al., 2021, Chen et al., 2024].

Constraints in the Dataset

This thesis is heavily bound by the data used to train the models and perform the evaluation. The DAIC-WOZ dataset is relatively small for fine-tuning deep transformer architectures with multiple attention mechanisms. This increases the risk of overfitting to training data.

Secondly, the dataset suffers from class imbalance, with most of the samples being healthy controls. Such an uneven distribution makes the training of the multiple regression heads for predicting rare symptom severity scores difficult.

Lastly, the data lacks structural standardisation. The interviews vary in total duration, token density, and overall conversational flow. Some patients elaborate deeply in their answers, while others give only minimal and defensive answers. This lack of standardisation introduces high variance into the text inputs, meaning that the predictability of their underlying diagnosis is heavily influenced by their communication style rather than their actual clinical state.

Self-Reporting Bias

The optimisation of the SMART model relies entirely on the training data, more specifically on the PHQ-8 scores from the patient questionnaires. However, these scores are self-reported by the patients, meaning they are subject to some degree of bias.

In clinical psychology, self-reporting bias is well known and occurs due to recall errors, social desirability, symptom exaggeration, or shame [Paulhus et al., 2007, Tourangeau and Yan, 2007]. During the conversational interview, especially when interacting with a virtual interviewer, patients often talk about their symptoms much more openly and accurately than on numeric questionnaires. This can result in large inconsistencies. The model may analyse the text and correctly identify linguistic markers of depressed mood. However, if the patient gave a "0" on their questionnaire for that symptom, the loss function penalises the model for its correct linguistic understanding during training. The model is essentially forced to throw away accurate predictions in favour of inconsistent and noisy human self-reports.

Semantic Over-Generalisation

Our SHAP analysis revealed another limitation regarding how the model learns the meaning of specific words. Because the different symptoms of depression usually co-occur in the training data, the model tends to overgeneralise vocabulary. For instance, our results showed that the isolated token *tired* strongly triggers emotional dimensions like Mood and Self-esteem, while the symptom head responsible for Energy remains largely unaffected.

This proves that the model did not learn the exact clinical meaning of physical fatigue. Instead, it treated a common symptom as a direct proxy for global depression. Consequently, the model may overestimate the severity of depression whenever a patient uses the word *tired* to describe depression unrelated to physical exhaustion.

Open Hypotheses

In our discussion on the layer-wise probing results (see Section 6), we formulated several hypotheses regarding how the multi-task objective influences the internal representation of language across the hidden layers. For example, we argued that the significant decrease of $\mathcal{V}$-usable information for our cognitive proxies (surprisal, self-reference, absolutist words) in the SMART model indicated an abstraction process. Token frequencies are translated into high-level clinical representations rather than simply being thrown away.

While the probing metrics provide reliable evidence regarding the availability of this information, the probing mechanism is a black box itself. It proves where the target information is encoded within the layers, but it cannot explain the exact internal processes or interactions. Due to the limited scope and timeline of this thesis, our architectural explanations still remain hypotheses that are yet to be definitively evaluated.

Future Work

Multimodal SMART Architecture

To overcome the modality gap and improve performance on somatic symptom heads like Energy and Psychomotor changes, future research should extend our SMART model into a fully multimodal architecture. The BERT encoder backbone should be paired with other encoders that specialise in non-textual features.

This could involve acoustic features using audio transformers such as Wav2Vec2 to capture speech rate or intonation. Simultaneously, vision networks like ResNets or visual transformers could detect visual signals like facial changes, eye movements, or posture. By implementing an additional cross-modal attention mechanism, the model could weigh text against audio or visual inputs. For instance, if a patient says "*I feel fine*", but the acoustic encoder detects a flat, exhausted voice and the visual encoder recognises a lack of facial expression, the multimodal attention could ignore the positive text and use the physical signals to make more accurate predictions in the somatic symptom heads.

Smarter Segment Aggregation

We identified that our current aggregation approach when moving from segment to patient-level predictions is suboptimal and can lead to significant signal dilution. Future work must therefore replace this segment-level mean aggregation with individually trained pooling layers. This could be implemented with a hierarchical attention network such that a secondary attention mechanism is employed on the segment embeddings and automatically learns the clinical relevance of each interview segment. This would allow the model to assign high attention weights to segments containing clear evidence of depression symptoms, while ignoring generic conversational fillers. Alternatively, the architecture could benefit from head-specific max-pooling layers for factual symptoms like Sleep or Appetite. This would ensure that if a severe symptom is only briefly spoken about in a single segment, the peak signal is fully preserved for the final patient-level score and not suppressed by the symptom scores of the other interview segments.

Contextualising Inputs with Interviewer Integration

Another possible direction of the SMART architecture could include the dialogue parts of the virtual interviewer in the model inputs to give more context and resolve semantic ambiguities. Instead of processing the patient segments alone, we would feed the model with question-answer pairs.

As previously mentioned, earlier research has already demonstrated the positive performance impact of incorporating the interviewer's speech during model training, particularly for short segments that lack context. However, it is important to note that there is some evidence arguing against using an interviewer's speech for model training, as this might cause the model to simply learn basic heuristics [Burdisso et al., 2024]. Therefore, more research is needed to investigate the exact effects of incorporating therapist's prompts in automated depression detection.

Time-series Severity Prediction

Our UMAP projections demonstrated that our proposed architecture internally organises patient embeddings along a continuous severity gradient, rather than building discrete clusters. This continuous topology would allow for tracking the patient's position on this latent severity line over time. By analysing transcripts from consecutive therapy sessions, the model could visualise whether a patient's condition is improving or worsening. This would provide clinicians with a highly sensitive monitoring tool for the treatment progress and identify warning signals of relapse early on.

Attention Analysis BERTviz

We already acknowledged the limitations of our layer-wise probing experiments and the resulting open hypotheses. Future work should evaluate our open hypotheses. For example, this could be done with visualisation techniques like BERTviz [Vig, 2019] that give a detailed view of the attention matrices within the symptom-specific heads.

Such experiments could explicitly trace how different linguistic markers, such as absolutist words and first-person pronouns, as well as additional unexplored cognitive proxies, link to specific negative clinical targets across the individual layers. This analysis would explain how the multi-task objective changes the model's internal language representation and, therefore, help gain a more transparent view of the exact decision-making process behind the predicted severity scores.

Model Application to Other Disorders

The architecture of the SMART model is modular and not restricted to just depression classification or the PHQ-8 framework. A promising direction for future research is to investigate the transferability of our multi-task symptom prediction architecture to other mental disorders.

The network could be adapted to other mental health conditions by swapping the eight PHQ-8 symptom heads for different symptom criteria. For instance, the model could be trained on anxiety datasets using the GAD-7 dimensions [Spitzer et al., 2006] or on psychosis detection using the PANSS framework [Kay et al., 1987]. Testing the model against different clinical questionnaires would evaluate its capacity to learn generalisable symptom-level boundaries within computational psychology more broadly.

Active Learning and Dynamic Interviews

Finally, a promising future application is integrating the SMART architecture into a dynamic, real-time interviewing system. Since the model predicts the severity score for all eight symptoms simultaneously, it could evaluate an ongoing interview in real-time. If the model detects high uncertainty or a complete absence of linguistic signals for a specific symptom head (such as the somatic ones), it could send a feedback trigger to the virtual agent. The virtual interviewer could then dynamically change the conversational direction to ask targeted and personalised follow-up questions about that specific missing dimension. This interactive loop would actively prevent signal sparsity during the data collection and ensure standardised and symptom-rich transcripts for every patient.

Out-of-Domain Testing and Multilinguality

To address the validity concerns raised by [Patapati et al., 2025], future work should test the model in a different environment from the training environment. Evaluating the architecture on, e.g., datasets from social media would verify whether the model has learned robust linguistic markers rather than dataset-specific statistics. Furthermore, our models have only been trained in English. An obvious next step would be the transition to multilingual variants to investigate if the multi-task symptom predictions hold true across other languages and cultural expressions.

Chapter 8

Conclusion

The primary objective of this thesis was to bridge the gap between computational NLP and clinical psychology. Traditional binary classification models for depression often act as unexplained black boxes that suffer from feature collapse, learning simple linguistic heuristics rather than building a clinically grounded understanding of depression markers. To overcome this issue, we proposed SMART, a multi-task learning architecture that replaces the standard binary classification approach with symptom-specific attention heads to predict the eight severity scores of the PHQ-8 framework together with a global binary diagnosis. Our evaluation demonstrated that shifting from a binary classification objective to symptom prediction changes how a language model structures and processes information. We successfully prevented feature collapse that caused the single-task baselines to completely fail. Our SMART architecture achieved a high performance, successfully identifying 81% of depressed patients. Since the final model diagnosis is based entirely on the summed predicted symptom severity scores, rather than relying on a direct binary output, this proves that by forcing the model to focus on distinct clinical markers, it builds a robust internal representation that is more promising than the traditional training objective.

Additionally, the model acts more human-like by classifying depression from individual symptoms, mimicking the diagnostic process of a human clinician. Beyond predictive accuracy, our explainability analyses highlighted the structural benefits of multi-task symptom learning. The latent space projections of the [CLS] tokens revealed that our architecture does not just put patient embeddings in distinct clusters but maps them to a continuous severity gradient. Furthermore, layer-wise probing confirmed that the multi-task objective changes how the network processes psycholinguistic features such as absolutist words or self-reference. The SMART architecture transitioned from encoding word counts in early layers to learning high-level representations in its final layers. The symptom heads learned realistic clinical correlations between the symptoms, proving that the model successfully captured the interconnectivity of depression.

Further token and error evaluations also exposed the strict boundaries and limitations of unimodal, text-based models. While cognitive and emotional symptoms like Mood or Self-esteem can be accurately extracted from transcripts, somatic symptoms like Energy or Psychomotor skills do not manifest as often in language, which leads to a modality gap. Additionally, using SHAP, we found that models underlie a risk of semantic over-generalisation, where the encoder learns to use physical words like *tired* as global proxies for depression instead of understanding their clinical meaning. These limitations highlight that text alone is insufficient for complete clinical plausibility.

Ultimately, this thesis provides strong evidence that the future of computational mental disorder diagnosis should move away from binary or multi-class classification. By integrating symptom-specific multi-task learning and robust explainability methods, we can develop models that are not only more accurate, but also transparent, interpretable and clinically grounded. The SMART architecture is one promising foundation for building multimodal and highly transparent monitoring systems for mental health.

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